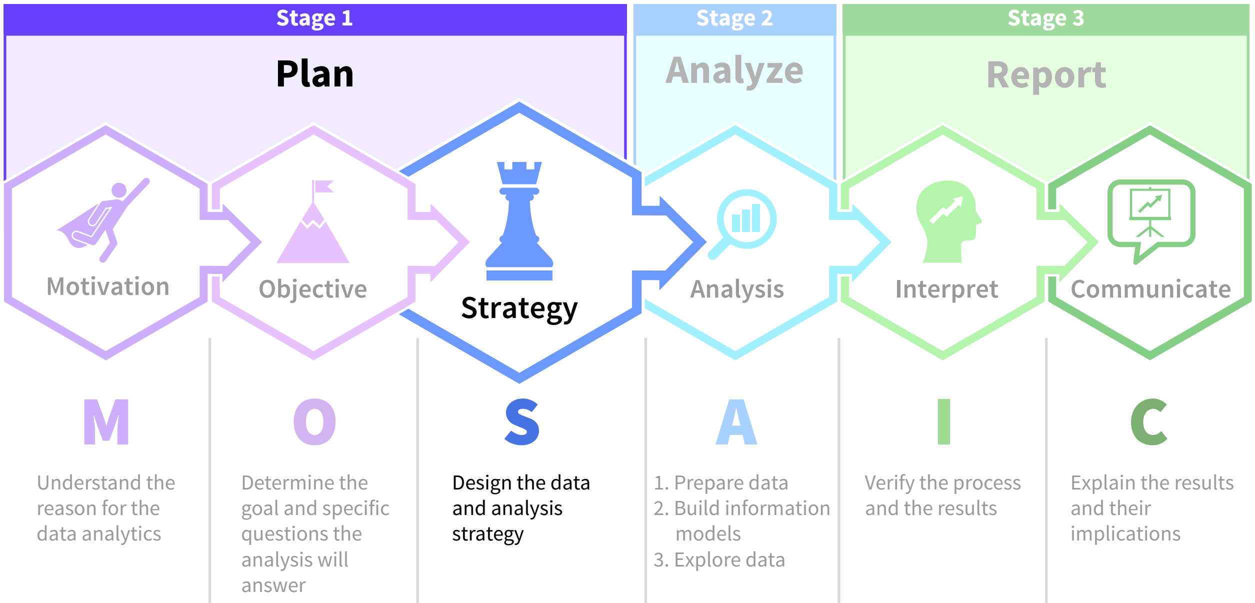
CHAPTER 4

Planning Data and Analysis Strategies

Chapter Preview

Armed with an understanding of the motivations and objectives for the data analysis project and after formulating the necessary objective questions, it is time for the final step in the planning stage–designing the data and analysis strategies. You will soon be expected to apply your knowledge, common sense, and problem-solving skills to design a variety of accounting, audit, and tax related data and analyses strategies. This chapter will help you develop the skills necessary to create effective data analysis project plans.



Professional Insight: Why Should You Plan Data and Analysis Strategies?

Taylor was excited to begin her career in auditing after interning at a Big Four accounting firm during the last busy season.

> My audit team had me design my own data and analysis strategies and document each step I performed. While I lacked confidence at first, I felt very supported. I learned so much, even when I made mistakes and had to pivot my strategies.
>
> **I was more successful when I understood the motivation for and the objective of my tasks. I was able to formulate better questions, which resulted in better**

data choices. The measurement scales of my selected data also pointed me toward the appropriate analyses.

My first task was a descriptive analysis. I had to determine if the year-end accounts receivable subsidiary ledger and general ledger balances reconciled. After checking if any credit customer's activities were excluded from the general ledger accounts receivable balance, I created an accounts receivable aging report listing each credit customer's outstanding balances from the subsidiary ledger. I compared the aging reports' items to the general ledger accounts receivable postings and noted each unmatched item. If there were no unmatched items, the accounts receivable balance in the general ledger was successfully reconciled and verified.

I then designed a diagnostic analysis strategy to determine which receivable activities were causing any unmatched items I had found. My strategy began with gathering the client's accounts receivable process documentation. For each unmatched item, I identified which process caused the items to be included in both ledgers. Documenting my work helped me explain what, how, and why I performed each step. This record will be helpful when we repeat this verification in next year's audit. **Designing my own data and analysis strategies helped me gain relevant skills and the confidence to start my auditing career.**

Chapter Roadmap

LEARNING OBJECTIVES	TOPICS	APPLY IT
LO 4.1 Identify the components of a data analysis project plan.	• Create a Data Analysis Project Plan • Sample Data Analysis Project Plan	**Build a Project Plan for Inventory Costs** (Example: Financial Accounting)
LO 4.2 Describe how to develop a data strategy.	• Identify Appropriate Data • Evaluate Data Fields and Sources • Consider Data Strategy Risks and Implement Controls	**Identify Data Characteristics** (Example: Accounting Information Systems)
LO 4.3 Explain how an analysis strategy is designed.	• Designing Analyses to Describe and Diagnose • Designing Analyses to Predict and Prescribe	**Create a Predictive Data Analysis Project Plan** (Example: Financial Accounting)
LO 4.4 Summarize data and analyses strategies in professional practice areas.	• Accounting Information Systems • Auditing • Financial Accounting • Managerial Accounting • Tax Accounting	**Match Strategies with Professional Practice Areas**

Data The Data tag appears in the chapter when the data for an example, illustration, or application are available on Wiley's online learning platform.

Data analytics software is continuously changing, and there may be more recent versions of the software referenced in this chapter. For more information access the accompanying video on Wiley's online learning platform.

4.1 How do Accountants Design Data Analysis Projects?

LEARNING OBJECTIVE 1

Identify the components of a data analysis project plan.

Recall that the first step in planning a data analysis project is understanding the motivations for the project, such as identifying problems, evaluating patterns and opportunities, measuring performance, and regulatory compliance. Considering the perspectives of the project's primary stakeholders is an important part of that understanding. Next, thinking critically about a project's purpose helps define the objective and articulate the project's specific questions. Common objectives and questions can be grouped by whether the analysis results describe, diagnose, or predict an outcome, or if the analysis results should prescribe future organizational strategy.

The final step of the data analysis planning stage is designing a professional plan for the data analysis project. Critical thinking continues to play a vital role, from selecting a strategy for the data and the analysis, to identifying inherent risks to both and embedding internal controls to reduce them. A data and analysis strategy plan is an organized and deliberate blueprint for the project. It also allows someone other than its creator to perform the analysis, which can free accountants to perform other valuable services with their business and regulatory expertise.

Create a Data Analysis Project Plan

Many of us use planning tools every day, such as relying on navigation applications to plan travel time and how to get to new destinations. Planning tools make tasks more efficient and the results more effective. They also often provide logical explanations for the necessity of certain steps or why they should be performed in a particular order.

This chapter summarizes a tool for data analysis project planning that provides similar benefits for accountants (**Illustration 4.1**). The order of the components is intentional:

Step 1 Focus on the objective:

- Keep the project's objective and specific questions in mind to select the best data and analysis strategies to fulfill the objective and answer those questions. Simply remembering to ask how the plan's proposed data and analysis strategy decisions relate to the objective helps us make better choices.

Step 2 Select a data strategy:

- Use critical thinking to develop and rank a few data alternatives. This ensures that we choose the data option most appropriate for the objective.

Step 3 Select an analysis strategy:

- Use what was learned from selecting the data strategy and apply that same development and ranking process to analyses alternatives. Following this step increases the likelihood of selecting the best analysis option.

Step 4 Consider risks:

- Considering and prioritizing critical risks to both the data and the analysis strategies reveals how these risks can create misleading and invalid results.

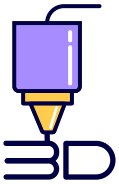
Step 5 Embed controls:

- Designing and implementing preventative and detective controls into the analysis process leads to results that are accurate, valid, and reliable.

ILLUSTRATION 4.1 A Data Analysis Project Plan

A data analysis plan increases the likelihood that analysis results accurately describe what happened, diagnose why it happened, predict what is likely to happen, or prescribe the best course forward. When the project plan is complete, so is the data analysis process planning stage, and it is time to move to the analysis stage. The next chapter describes how to properly prepare data for analysis. But first, practice using this project planning tool in an accounting context.

Sample Data Analysis Project Plan



Imagine you are a financial accountant for WeMakeIt, Inc., which is an employee-owned 3D printing corporation in Phoenix, Arizona. They sell superhero figurines, mechanical shop tools, and medical prostheses that are manufactured with 3D printing technologies. Founded five years ago by a local college student, Marisabel Cordoba, WeMakeIt, Inc. grew quickly to fill their production capacities and has maintained a stable size for the last three years.

Step 1: Focus on the Objective

The project in this example is to estimate the bad debts in the 2025 year-end accounts receivable balance. The result will be the value estimate for the 2025 year-end adjusting journal entry that debits bad debts expense and credits the allowance for uncollectible accounts contra-asset.

APPLYING CRITICAL THINKING 4.1: Estimate Bad Debts

The U.S. GAAP ending accounts receivable method requires the following:

- The year-end accounts receivable balance can be considered in total or in age group subtotals. Age grouping rules for manufacturing companies are typically less than 30 days, 30 to 60 days, and greater than 60 days outstanding, but these vary by industry.
- Reviewing regulatory guidance, industry practices, business policies and processes, and past bad debt estimation percentages is the professional approach for selecting bad debt estimation percentages. Typically, as the age of the accounts receivable increases, the percentage uncollectible applied to that age also increases **(Knowledge)**.

Assume the following for this example:

- The company offers its credit customers payment terms of 30 days.
- The 2025 year-end accounts receivable balance in the general ledger is $83,734.30.
- The 2025 year-end unadjusted allowance for uncollectible accounts has a credit balance of $3,000.
- WeMakeIt, Inc. decided to use the percentage of year-end accounts receivable method to estimate bad debts expense each year. They have authorized a 2% bad debts assumption for invoices that are 30 to 60 days outstanding and 30% for invoices that are outstanding for more than 60 days.

This data analysis project has the following objective and related questions:

- **Objective:** Accurately estimate bad debts within the year-end accounts receivable balance in accordance with U.S. GAAP.
- **Specific questions:**
  1. What is a reasonable estimate of the uncollectible receivables in the 2025 year-end outstanding accounts receivables?
  2. What amount should be used in the adjusting journal entry for bad debts expense?

With the objective defined and the questions articulated, it is time to develop the data strategy.

Step 2: Select the Data Strategy

Develop several data alternatives that could help answer the objective question. Then, to select the most useful data alternative for the project plan, identify the factors you want to use to rank these alternatives, and assign values to each alternative's factors. The best data strategy alternative is the one with the highest overall factor ratings.

APPLYING CRITICAL THINKING 4.2: Consider Different Perspectives

The net realizable value of accounts receivable (accounts receivable minus the allowance for uncollectible accounts) is reported on the balance sheet as a current asset. The bad debts expense reduces net income reported on the income statement. Keep in mind that parties with very different perspectives use this information for decision-making **(Stakeholders)**:

- Creditors, who want to be repaid, are interested in the liquidity of the assets.
- Shareholders, who predict future stock values as well as dividend expectations, must understand the expected future cash inflows from operations.
- Managers want to earn bonuses that are determined by growth in net income and total assets. The change in the allowance for uncollectible accounts will influence both amounts.

There are a few data strategy alternatives that could be considered either individually or in combination for this project:

- All credit sales invoices and collections data that pertain to the accounting period.
- All outstanding invoices included in the year-end accounts receivable balance.
- The prior year-end's adjusted account balance and the current year-end's unadjusted account balance in the allowance for uncollectible accounts.
- Current year's total accounts receivable write-offs.
- Authorized bad debt estimation percentages for accounts receivable age categories.

Recall that the objective question for this analysis project asks for a reasonable estimate of the year-end uncollectible accounts receivables. Because WeMakeIt, Inc. uses the GAAP percentage of ending accounts receivables method with the three age categories typical for manufacturing companies, the data that would be most useful considering the objectives would include:

- The outstanding invoices in the year-end accounts receivable balance.
- This year's unadjusted allowance for uncollectible account balance.
- Authorized bad debt estimation percentages for each accounts receivable age group.

APPLYING CRITICAL THINKING 4.3: Choose the Data Strategy that Fits Your Objective

Both data and analysis strategies are improved when they are developed, evaluated, and ranked by the factor values placed on each potential data strategy:

- Factors that value data relevance, such as expected future collections, prior estimation errors, and expected risk of uncollectibility should be used to select the best data strategy for WeMakeIt, Inc's analysis plan **(Alternatives)**.
- Using the data from outstanding invoices, the unadjusted allowance for uncollectible accounts receivable, and authorized bad debt estimation percentages was the best choice for WeMakeIt, Inc's analysis. These data choices best match the objective of the analysis, which is to calculate a reasonable estimate for the year-end uncollectible accounts receivable **(Purpose)**.
- For a personal example, if your objective was to dress successfully for an interview, then you could develop, evaluate, and rank the alternatives of wearing something you already own, borrowing from a friend, or buying something new to wear to the interview. Using clothes you already own would rank higher than buying new clothes if you were ranking options based on lowest costs and time factors.

Step 3: Select the Analysis Strategy

For any project plan, choosing the best analysis strategy involves considering and evaluating several possible alternative analyses given the objective questions and the already selected strategy for the data. The analysis alternatives in this bad debt estimation example involve different methods for determining the age of an outstanding invoice:

- Days outstanding from the invoice date forward.
- Days outstanding from the due date forward.

The first analysis alternative is preferable because WeMakeIt, Inc.'s policy is to allow customers 30 days to pay their invoices and because they estimate bad debts only when an invoice is past its due date, which is equal to or more than 30 days outstanding.

Therefore, the first question, what is a reasonable estimate of the uncollectible receivables in the 2025 year-end outstanding accounts receivables, will be answered by summing

How To

each age group's estimated bad debts value. This is a measure estimating the uncollectible portion of the year-end accounts receivable balance (Illustration 4.2). (Data See How To 4.1 at the end of the chapter to learn how to calculate the figures in this illustration.)

ILLUSTRATION 4.2 WeMakeIt, Inc.: Estimated Value of Uncollectible Receivables

	< 30 days	30–60 days	> 60 days	Total
Total accounts receivable value	$ 30,099.29	$ 20,441.46	$ 33,193.55	$ 83,734.30
x % uncollectible	0%	2%	30%	
$ Estimated as uncollectible	$ 0	$ 408.83	$ 9,958.07	$ 10,366.89

The second question, what value should be used in the adjusting journal entry at year-end for bad debts expense, will be answered when this total of $ 10,366.89 is reduced by the unadjusted year-end allowance for uncollectible accounts balance ($ 3,000). The resulting calculated value is the year-end adjusting journal entry debiting bad debt expense and crediting allowance for uncollectible accounts:

	Debit	Credit
Bad debts expense	$ 7,366.89	
Allowance for uncollectible accounts		$ 7,366.89

Increasing the allowance account's $3,000 unadjusted balance by $ 7,366.89 will result in an adjusted balance of $ 10,366.89, equal to the estimated uncollectible amounts in accounts receivable at the 2025 year end. Illustration 4.3 summarizes the first three steps of this project's plan.

ILLUSTRATION 4.3 In-Progress Project Plan for Estimating Year-End Bad Debts: Steps 1-3

1 Objective Use U.S. GAAP to estimate bad debts in year-end accounts receivable balance.

Questions: 1. What is the estimated value of uncollectible accounts receivable at year-end?
2. What is the amount that should be used in the year-end bad debts adjusting journal entry?

2 Data Strategy

1. Outstanding invoices in year-end accounts receivable.
2. Authorized estimated bad debt percentages: 2% for 30–60 days; 30% for > 60 days groups.
3. Unadjusted allowance for uncollectible accounts year-end balance.

3 Analysis Strategy

1. Age outstanding invoices by sorting by date into < 30 days, 30–60 days, and > 60 days groups.
2. Create a total of the invoice values for each of these three aged groups.
3. Multiply 30-60 days group total by 2% for the uncollectible value estimate for this group.
4. Multiply > 60 days group total by 30% for the uncollectible value estimate for this group.
5. Sum uncollectible value estimates from steps 3 and 4 for total uncollectible value estimate.
6. Subtract step #5 total from Allowance for Uncollectible Accounts unadjusted balance of $3,000 for the adjusting entry value.

This total answers the first project plan question

This result answers the second project plan question

4 Data Strategy Risks | **Analysis Strategy Risks 4**

5 Data Strategy Controls | **Analysis Strategy Controls 5**

This project plan is incomplete. The steps for evaluating risks and embedding controls are covered next.

The last two steps are considering risks and controls for both the data strategy and the analysis strategy. First we will evaluate the data strategy's risks and controls, and then the analysis strategy risks and controls.

Steps 4 and 5: Data Strategy Risks and Controls

While organizations invest in processes and controls designed to capture accounting data correctly, accounting databases inevitably have some data issues. In this example, the data include the 2025 year-end accounts receivable outstanding invoices and the authorized bad debt estimation percentages. There could be some risks to using this accounts receivable data:

- Missing or incorrectly included invoices in the general or the subsidiary ledger.
- Errors in the invoice due date fields or the invoice total amounts.
- Errors created when the data are exported from the AIS system into the analysis tool.

APPLYING CRITICAL THINKING 4.4: Threats to Data Strategy

Data risks can impact the accuracy and validity of analysis results. These risks must be evaluated so appropriate controls can be added to the project plan. Data strategy risks can include **(Risks)**:

- Dirty data such as incomplete data, inaccurate data, and incorrectly formatted data fields.
- Data irrelevant to the objective of the project.
- Insufficient data for the analysis to be reliable.
- Data that is an unrepresentative sample of the underlying population.
- Errors in management estimates and assumptions about the data.

These data risks can be controlled by comparing the extracted data to the source invoice and collection documents. Possible risks involving the management's estimates of bad debt percentages include human bias embedded in the authorized bad debt percentages, changes in customer payment behavior, and business process changes to credit customer approvals and collection practices.

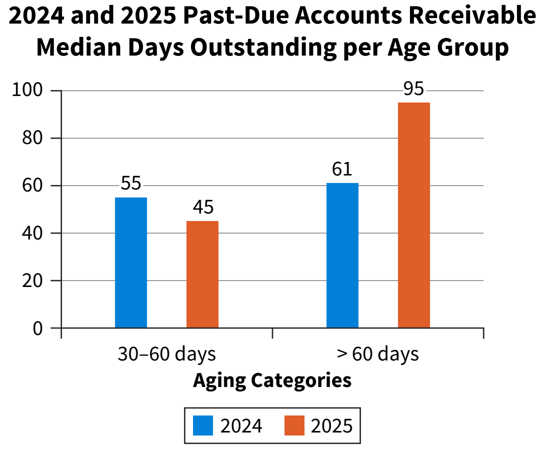
One way to control for these risks is to ask the finance, sales, and accounts receivable managers if there were changes to the customer base, market conditions, or business policies and procedures for credit approval, write-offs, or collection policies during the year. This knowledge could confirm existing bad debt percentages or motivate their adjustment.

Another control for evaluating the risks in management estimates and assumptions is to compare current data to prior year data. Evaluating increases in the median number of days outstanding and the number of new to returning customers in each age group can offer insights into the reasonableness of bad debt estimation percentages.

To illustrate, imagine you were wondering if the 30% bad debt percentage for invoices greater than 60 days outstanding was a reasonable management assumption in 2025.

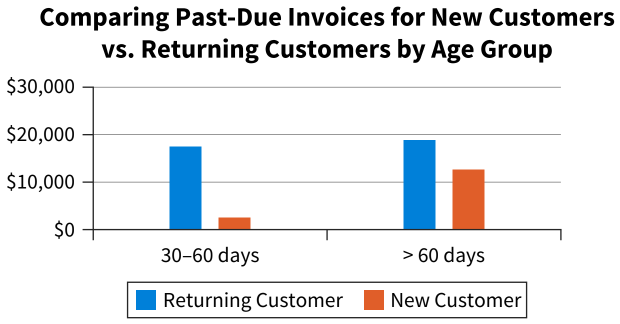
Illustration 4.4 demonstrates that the median number of days outstanding in the greater than 60 days outstanding group increased in 2025.

ILLUSTRATION 4.4 Data Risk in Bad Debt Percentages: Median Days Outstanding



Additionally, you might wonder if this increase was caused by new customers, as the payment habits of new customers are more uncertain than those of continuing customers. **Illustration 4.5** shows that the > 60 days age group has many outstanding invoices from new customers.

ILLUSTRATION 4.5 Data Risk in Bad Debt Percentages: Risks from New Customers



The results of these control tests indicate that if the 2024 bad debt percentage for this age group was used in 2025, it may not accurately measure the bad debts expected in the 2025 accounts receivable balance. These visualizations show that implementing controls, such as data reasonableness tests, before performing an analysis can result in better business and professional intelligence.

Steps 4 and 5: Analysis Strategy Risks and Controls

Common analysis risks include errors in data manipulation, calculation, or using technology, as well as human biases and ethical violations when using sensitive data. Here are some examples of analyses risks when estimating bad debts:

- Errors in age group categorization and age total calculations.
- Incorrect application of bad debt estimation percentages to age group totals.
- Erroneous summing of estimated bad debts across age groups.
- Mistakes in the calculation of the value to use for bad debts expense.

Verifying the accuracy of the age group categorizations and each sum and multiplication performed on age group data items and totals would control for these risks.

Another control is to use common sense when reviewing the results for reasonableness. **Illustration 4.6** uses an example from the 30–60 days age group estimation to show how a common sense check helps catch large errors. The total outstanding accounts receivable for that age group was $ 20,441.46, and the uncollectible percentage for that group was 2%. The estimated uncollectible value for that age group was calculated as the product of those two numbers, resulting in $ 408.83. This control has verified our calculation.

ILLUSTRATION 4.6 Common Sense Check for the 30–60 Days Group Results

Common Sense Step 1	Common Sense Step 2	Common Sense Step 3
2% is 1/5th of 10%	10% of $20,441 = $2,044	$2,044 divided by 5 = $409
10% is easy to calculate!	Now, just divide this by 5 to get the 2% amount.	Results for this age group were $408.83. We can be confident in our result.

Analysis results that significantly vary from common sense checks should be verified.

The project plan design for this example is now complete (**Illustration 4.7**).

ILLUSTRATION 4.7 Complete Project Plan for Estimating Bad Debts

1 Objective Use U.S. GAAP to estimate bad debts in year-end accounts receivable balance.

Questions: 1. What is the estimated value of uncollectible accounts receivable at year-end?
2. What is the amount that should be used in the year-end bad debts adjusting journal entry?

2 Data Strategy The following data should be used in the analysis:

- Outstanding invoices in year-end accounts receivable.
- Authorized estimated bad debt percentages: 2% for 30–60 days; 30% for > 60 days groups.
- Unadjusted allowance for uncollectible accounts year-end balance.

3 Analysis Strategy

1. Age outstanding invoices by sorting by date into < 30 day, 30–60 days, and > 60 days groups.
2. Create a total of the invoice values for each of these three aged groups.
3. Multiply 30–60 days group total by 2% for the uncollectible value estimate for this group.
4. Multiply > 60 days group total by 30% for the uncollectible value estimate for this group.
5. Sum uncollectible value estimates from steps 3 and 4 for total uncollectible value estimate.
6. Subtract step #5 total from allowance for uncollectible accounts unadjusted balance of $3,000 for the adjusting entry value.

This total answers the first project plan question

This result answers the second project plan question

4 Data Strategy Risks

- Data errors (missing data, due date errors, invoice total errors).
- Data extraction errors.
- Inaccurate management assumptions and estimates (bad debt percentages).

Analysis Strategy Risks 4

- Errors in age group categorization and calculations.
- Erroneous application of bad debt estimation percentages to age group totals.
- Erroneous summing of estimated bad debts across age groups.
- Errors in the calculation of the value to use for bad debts expense.

5 Data Strategy Controls

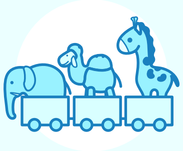
- Compare extracted data to source documentation.
- Compare to prior period to verify management estimates and assumptions.
- Review customer, policy and procedure changes.

Analysis Strategy Controls 5

- Verify all classifications and groupings.
- Verify all calculations.
- Verify correct use of analysis technologies.

Next, we go into more depth about how to develop both data and analysis strategies.

Apply It 4.1
Build a Project Plan for Inventory Costs



Financial Accounting Founded by Elisabeth Hess and based in Wisconsin, Tremendous Toys produces and sells unique animal-form toys online. The company has over a million dollars in sales revenues and a strong national and international following. The investment bank underwriting its upcoming initial public offering requires that Elisabeth produce a U.S. GAAP valuation of year-end inventory.

Elisabeth has kept meticulous records, organized in business cost categories, which are listed here. She knows which materials were used to create each toy, which employee built it, and how long each took to create. She has also recorded the number and type of toys produced every month. However, she has never assigned costs to her inventory items.

Materials	Labor	Overhead
Plastic for 3D printers	Production workers' time	3D printer costs
Wood	Production workers' pay rate	Production tools costs
Resin		Computer (peripherals and internet) costs
Fiber and fabric		Shipping costs
Paints, varnishes, and dyes		Factory rent, utilities, and storage
Toy clothing		Production supplies
Glues and epoxies		Tax and legal fees
Bling and other accessories		Donations to charities
Eyes and body parts		Advertising costs

Prepare a data analysis project plan to help Elisabeth develop her inventoriable costs. Use the following structure for your answer.

1. Project objective and specific question
2. Data strategy
3. Analysis strategy
4. Data and analysis strategy risks
5. Data and analysis strategy controls

SOLUTION

1. Project objective	Prepare a U.S. GAAP compliant full costing valuation for Tremendous Toys' current inventory.
Specific question	What is the U.S. GAAP year-end value of inventory?
2. Data strategy	• Collect all costs and raw measures of consumption of resources related to production (direct materials, direct labor, and manufacturing overhead costs) for the last three years. • Collect records of how many units were produced for each month of cost data received. • Take a physical count of the current inventory by product.

(Continued)

3. Analysis strategy	• Aggregate cost data by monthly production of figurines. • Estimate cost behavior using regression to find variable costs per figurine and total fixed production costs per month. • Apply results to year-end inventory.
4. Risks to data strategy	• Missing data. • Including personal expenses. • Material costs and worker costs may not be representative of true replacement costs of future material and labor costs. • Potential bias for minimizing cost information to make business seem more profitable.
Risks to analysis strategy	• Averaging costs across figurines may not be accurate if true costs of different figurines vary significantly. • Regression may not have strong explanatory power (*R*-square). • Costs may have seasonality differences (due to scarcity of materials or heating and cooling costs). • Cost formula may need to be recalculated annually to capture cost changes.
5. Data strategy controls	• Interview Elisabeth with a typical list of inventory costs to catch omitted costs. • Inquire about the seasonal or economic trends of materials, labor and production overhead costs, and use the average of the highest and lowest points.
Analysis strategy controls	• Use median tendencies rather than means. • Run sensitivity models with future replacement cost values to note potential risk in final product cost estimations and possible data bias.

4.2 What Should We Consider When Selecting Data for Analysis?

LEARNING OBJECTIVE 2

Describe how to develop a data strategy.

It's easier to make better decisions when we have the necessary information. Each semester, students who know their educational and career goals can select the courses that will prepare them to graduate and begin their careers. For example, accounting majors do not need to take the same courses as education majors. The same idea applies when developing a strategy for selecting data in data analysis. A successful data analysis project hinges on selecting data that are relevant and appropriate for the objective of the project, respecting the data's characteristics and measurement scales, and controlling for inherent data risks.

Identify Appropriate Data

Data can be considered appropriate for analysis when they are relevant, available, and the characteristics match the analysis method requirements. Appropriate data can be internal, external, or a combination of both:

- Internal data are generated within the organization, such as sales data, purchase data, inventory data, customer data, and vendor data. Internal data can typically be more easily controlled and verified by an organization.
- External data are obtained from sources outside of an organization. This data can include weather data, geographic data, and publicly available competitor data. External data are somewhat riskier to use since we often cannot know if the data are accurate or complete. External data can, however, provide insights that internal data alone cannot provide.

After identifying the available and relevant data alternatives, the characteristics of the possible data sets need to be verified as suitable for the planned analysis.

Evaluate Data Fields and Sources

A **data set** is a collection of data columns and rows available for analysis. Understanding the characteristics of a data set is important because, for example, statistical measures and tests often require certain data characteristics or a minimum of data points. Violating the data requirements for these measures and tests can threaten the accuracy, reliability, and significance of the analysis results.

Let's use an example to evaluate the impact data can have on analysis. **Illustration 4.8** shows the inventory data set for the 3D manufacturing company WeMakeIt, Inc.

ILLUSTRATION 4.8 Typical Elements in a Data Set from WeMakeIt, Inc.'s Inventory

The data set is the entire collection of rows and columns.

Column = Data field = Attribute

Row = Record

InventoryCode	InventoryDescription	ProductCategory	UnitCost	NumberOnHand
358500	Medical model foot	1	$ 30.40	32
358600	Medical model hand	1	$ 30.20	16
358900	Shop Lamp globe	2	$ 20.05	43
359000	Cord wraps	2	$ 5.00	80
361100	Superhero Fantastic Four Mister Fantastic	3	$ 36.00	30
361200	Superhero Fantastic Four Invisible Woman	3	$ 36.00	35
361300	Superhero Fantastic Four Thing	3	$ 36.00	25
361400	Superhero Fantastic Four Human Touch	3	$ 36.00	30

The individual columns in a data set are called **fields**, and if the source of the data was a database, the columns are called **attributes**. Each column describes and represents one unique characteristic, a description, or aspect of the phenomena captured in the data set. Rows in a data set from a database are **records**, which represent the collection of columns that hold the descriptions of a single occurrence of the data set's purpose. In Illustration 4.8, the highlighted row contains all the information about the inventory item Superhero Fantastic Four Mister Fantastic–inventory code, the description, product category, unit cost, and the number of items on hand. This inventory item belongs to the ProductCategory group #3, indicating that it is a superhero figurine.

The records, or rows, in this inventory data set represent the different products manufactured in each product line. The columns that describe each inventory product for the WeMakeIt, Inc. inventory data set are listed in **Illustration 4.9**.

ILLUSTRATION 4.9 Data Dictionary for the WeMakeIt, Inc. Inventory Data Set

Field Name	Description
InventoryCode	Unique numerical code to identify inventory (primary key)
InventoryDescription	Text that describes the inventory product
ProductCategory	The type of product (1 = medical, 2 = tool, 3 = toy)
UnitCost	Currency value assigned to each unit's cost
NumberOnHand	Quantity of items in inventory

In addition to understanding the content of data fields in accounting databases, considering the source of the data is important because the quality of the data in the fields impact the quality of the analysis. Were the data created by an external party on a web page, entered internally by an employee, or automatically assigned by the computer? Do the data refer to a category, a measurement, or a calculation? For example, in the WeMakeIt, Inc. inventory data in Illustration 4.9, the InventoryCode field's data are probably automatic sequential numbers assigned by the AIS system whenever a new inventory product is added. Since a human is not creating that field's value when a new row is added, we can be confident that the InventoryCode data are consistently correct. **Illustration 4.10** summarizes the typical internal data field sources often used by accountants.

ILLUSTRATION 4.10 Common Data Field Sources and Examples

Source	Description	WeMakeIt Inc. Examples
Raw Data Fields		
Measured Raw Data	• Data created or captured by a controlled process capturing the value of the data. Examples include price, cost, number on hand, weight, date, hours worked, temperature, sensor readings, human observation, or economic value. • Their format can be discrete or continuous data.	InventoryCost NumberOnHand UnitCost
Non-measured Raw Data	• Data often created automatically by the computer or company policy for control. Examples include identification codes, chart of account numbers, standard descriptions, product category codes, or location codes such as city. • These fields are typically formatted as discrete data.	InventoryCode ProductCategory
Calculated Data Fields		
Calculated Data	• Data created when one or more fields in a particular record (row) have any number of mathematical operators (such as +, −, ×, %) applied, and often are derived from using the data in another field or fields within the same record (never across different rows). • These fields can be formatted as discrete or continuous data.	TotalCost (=UnitCost × NumberOnHand)

In the WeMakeIt, Inc. data set example, both the InventoryCode and ProductCategory fields' data are **non-measured raw data** that have been formatted as numeric and discrete, meaning that they are non-continuous and will never have partial unit values. These fields cannot be used in analysis for mathematical calculations because they refer to unique identifiers for inventory items and their product category group. The InventoryDescription field's data source is also a non-measured raw data field formatted as a text field, which cannot be used in numeric calculations during the analysis.

The sources for data fields like UnitCost and NumberOnHand are different because they are **measured raw numeric data**, which can be used in mathematical calculations.

Finally, UnitCost and NumberOnHand can be multiplied in an expression to calculate a new field within a query called TotalCost. This new field source is **calculated data** that is also numeric and likely to be used in a variety of different mathematical calculations in accounting.

A data field's **measurement scale** refers to the type of information provided by the data. Data measurement scales should be considered when designing the data strategy, as they impact which analyses can be performed on the data. Data measurement scales are grouped into four categories: categorical, ordinal, interval, or ratio (**Illustration 4.11**).

ILLUSTRATION 4.11 Measurement Scale Descriptions and Examples

Measurement Scale Classification	Description	Example
Categorical (nominal) data	Labeled or named data that can be sorted into groups according to specific characteristics. The data do not have a quantitative value.	Regions, product type, account numbers, account category, locations.
Ordinal data	Ordered or ranked categorical data. Distance between the categories does not need to be known or equal.	A survey question inquiring about customer service may ask customers to rate their experience from 1 to 10.
Interval data	Ordinal data that have equal and constant differences between observations and an arbitrary zero point.	Temperature, time, credit scores.
Ratio data	Interval data with a natural zero point. A natural zero point means that it is not arbitrary.	Economic data, such as dollars or euros.

We discuss how data measurement scales influence analysis strategy options later in this chapter. Next, let's examine the risks and controls related to our data strategy.

Professional Insight 4.2: How Do You Select Financial Data for Analysis?

Jadon is an accounting and finance major, student athlete, and president of the first-generation student organization on campus. During his internship, he realized how much his career would involve data analysis of financial statement information.

> **On the first day of my summer internship, I was tasked with calculating ten financial ratios for five companies in the same industry for the most recent three years.** I was left to figure out how to get the data and perform the calculations on my own. I reached into my sports experience to gather my courage by creating a game plan. **First, I had to select and set up my analysis tool.**
>
> I chose Excel with the Analysis ToolPak and made a raw data template based on the totals I would need for the ratios in the first worksheet. I copied it five times for each company's raw data. I made another worksheet for all the ratio calculations as the rows, and the companies would be the columns, grouped by each year. I organized the ratios into profitability, liquidity, and solvency groups. I entered ratio formulas that tied back to the raw data worksheet cells and documented my steps on another sheet to remember my choices.
>
> **Next, I had to find out where to get the data. I went to the SEC's xBRL data and downloaded the financial statement information and entered what I needed into Excel.** An unexpected benefit of having my ratios all in one sheet was that it was easy to make visualizations of my results. I was amazed by how clear the best and worst performing companies were from these visualizations. I saved my file and emailed it to my boss.

Consider Data Strategy Risks and Implement Controls

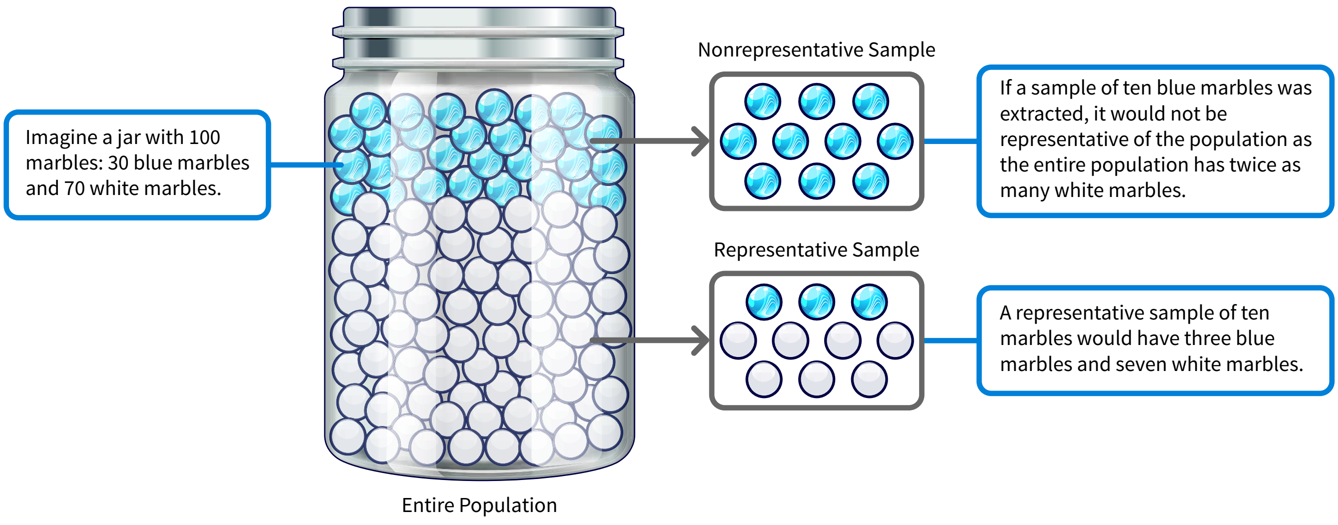
Another benefit of documenting data strategy in a project plan is that examining each choice can help identify three common data risks that might affect an analysis's validity, accuracy, and reliability (**Illustration 4.12**).

ILLUSTRATION 4.12 Common Data Risks and Recommended Controls

	Risks	Controls
Data	Non-representative sample selections	Verify representativeness of sample.
	Outlier data points	Perform a histogram or quartile analysis to identify outliers, then explain or justify the rule used for outlier adjustment or removal.
	Dirty data	Verify integrity of data set and clean up dirty data issues.

The first risk is that a sample extracted from a larger population of data are a poor representation of the underlying population. For example, a random selection of different colored marbles might not be a sample that accurately represents the characteristics of the marble population (**Illustration 4.13**).

ILLUSTRATION 4.13 Representative Sampling Risks



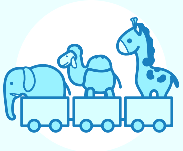
That is why performing tests on the sample's representativeness can either verify its validity or detect its significant weakness in representing the underlying marble populations.

The second risk is potentially including unusual data points in an analysis. Outlier data points are unusual data points compared to the rest of the data in either an activity point (x-axis or the independent variable), such as number of units produced, or an unusual level of the economic value (y-axis or the dependent variable), such as total costs or total sales. The best control for identifying outliers is to visualize the data in a graph to check if the measure is very different than the rest of the data. Identified outliers can either be remeasured (if possible) or removed with a logical and documented rule.

The final data risk is that a variety of data errors, or **dirty data**, are problematic for data analysis. Dirty data include missing, invalid, duplicate, and incorrectly formatted data. For example, the purchase orders, invoices, or checks written by the company should be sequentially accounted for, with no missing numbers or two with the same identifying number. Controls should always test for dirty data before the analysis is performed. Data can be compared to its source documents, or if that is not possible, tested for reasonableness. Testing for reasonableness can include checking for missing sequence numbers, duplicate numbers, or verifying the data have the expected format or acceptable characters.

Once data have been selected, evaluated, and any risks identified and controlled, the next step is to develop an analysis strategy appropriate for the objective of the project.

Apply It 4.2
Identify Data Characteristics



Accounting Information Systems Tremendous Toys has recently developed a database to capture inventoriable costs from their production each week.

Data has been extracted from the following data fields:

Field Name	Field Type	Field Measurement Scale
ProductionRunNumber		
ProductionRunDate		
NumberOfToysinRun		
ToyIDNumber		
DirectMaterialsUsed		
DirectLaborUsed		
OverheadCostApplied		
TotalRunCost		
TotalUnitCost		

Complete the three-column table by adding each data field's type and measurement scale.

SOLUTION

Field Name	Field Type	Field Measurement Scale
ProductionRunNumber	Non-measured raw data	Categorical (Nominal)
ProductionRunDate	Non-measured raw data	Interval
NumberOfToysinRun	Measured raw data	Ratio
ToyIDNumber	Non-measured raw data	Categorical (Nominal)
DirectMaterialsUsed	Measured raw data	Ratio
DirectLaborUsed	Measured raw data	Ratio
OverheadCostApplied	Calculated data	Ratio
TotalRunCost	Calculated data	Ratio
TotalUnitCost	Calculated data	Ratio

4.3 What Should We Consider When Selecting an Analysis?

LEARNING OBJECTIVE 3

Explain how an analysis strategy is designed.

Consider these two questions when designing a data analysis strategy:

1. Can the chosen analysis strategy answer the specific objective questions?
2. Is the measurement scale of the data appropriate for the selected analysis strategy?

Earlier in this course you learned there are four types of data analysis objectives: descriptive, diagnostic, predictive, and prescriptive. Each asks different questions of the data. We perform analyses to answer those questions. The specific analysis that can be performed on the data is influenced by the data's measurement scales. Next, we summarize what to consider when designing analyses for projects with descriptive and diagnostic objectives.

Designing Analyses to Describe and Diagnose

Because understanding the phenomena captured by the data is essential, many data analysis projects begin with descriptive analyses. Diagnostic analytics involve many of the same analyses as descriptive, but the focus is on explaining why something happened. Because many types of analyses can be both descriptive and diagnostic, it makes sense to discuss these two analysis strategies together.

The specific type of descriptive and diagnostic analysis used depends on the measurement scale of the data. For example, calculating the median observation in the data requires using either interval or ratio measures. **Illustration 4.14** shows some analysis strategies for descriptive and diagnostic objectives (checkmarks indicate appropriate analyses, and Xs indicate analyses that are inappropriate for the measurement scale).

ILLUSTRATION 4.14 Analyses Strategies for Descriptive and Diagnostic Analytics

	Data Measurement Scale			
Potential Analyses	**Categorical**	**Ordinal**	**Interval**	**Ratio**
Central Tendency				
Mode	✓	✓	✓	✓
Median	✗	✗	✓	✓
Mean	✗	✗	✓	✓
Frequency Distributions				
Both counts and %s:				
Group frequency	✓	✓	✓	✓
Cumulative frequency	✓	✓	✓	✓
Relative frequency	✓	✓	✓	✓

(Continued)

ILLUSTRATION 4.14 *(Continued)*

	Data Measurement Scale			
Potential Analyses	**Categorical**	**Ordinal**	**Interval**	**Ratio**
Data Dispersion				
Minimum, maximum, range, quartiles	✗	✓	✓	✓
Variance and standard deviation	✗	✗	✓	✓
Normality, skewness, and kurtosis tests	✗	✗	✓	✓
Visualizations				
Only bar and pie chart	✓	✓	✓	✓
Many other trends options	✗	✗	✓	✓
Histograms/box plots	✗	✓	✓	✓
Correlation				
Scatterplots, Pearson's R	✗	✓	✓	✓
Cross tabulation analysis	✓	✓	✓	✓
Calculations				
Totals and subtotals	✗	✗	✓	✓
Percent change	✗	✗	✓	✓
Percent of total	✗	✗	✓	✓
Regression	✗	✗	✓	✓

Descriptive Analysis Strategies

Recall that accounting data are historic economic transaction and valuation data. Accountants, business managers, and regulators all want to know more about what happened and whether the data signals good or concerning news, so descriptive analyses are the most common analysis strategies using accounting data. It is useful for evaluating strategy performance because it provides more meaning and business intelligence than from just looking, for example, at the totals reported on one year's financial statements.

Let's work through an example using the using the WeMakeIt Inc. accounts receivable data set to illustrate the explanatory power of descriptive analysis even when using categorical data. Recall that the objective for this analysis was to become more confident in the percentages used for estimating bad debts. The initial descriptive analysis results indicated a large increase in the number of customers with open invoices at the 2025 year end compared to the 2024 year end. This analysis had been designed to answer two descriptive questions:

- **Objective question 1:** How many customers had outstanding invoices at the year end of 2024?
- **Objective question 2:** How many customers had outstanding invoices at the end of 2025?

A data strategy and an analysis strategy were developed:

- **Data strategy:** Use the categorical variable of CustomerNumber to identify customers with outstanding balances.
- **Analysis strategy:** Count and compare the frequency of unique customer numbers with outstanding invoices at the year-end of 2024, and then repeat the analysis for the 2025 year-end. Note that both count and frequency are analyses that can be performed on categorical data.

Illustration 4.15 reports the results of this analysis.

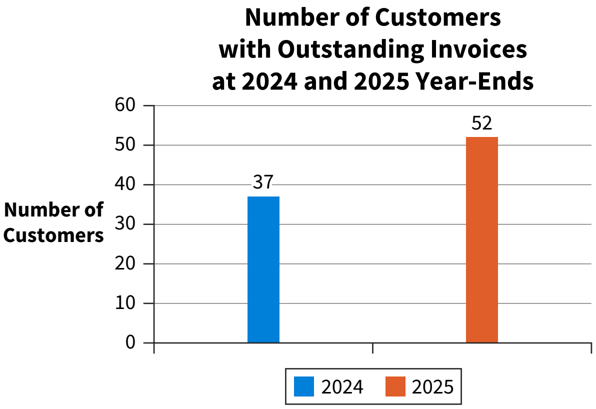


ILLUSTRATION 4.15
Outstanding Invoices by Customer Number

The results indicated that in 2024 there were 37 unique customer numbers with outstanding invoices, and in 2025 there were 52 unique customers numbers with outstanding invoices. Next, we discuss diagnostic analysis strategies.

APPLYING CRITICAL THINKING 4.5: Be Careful Interpreting Initial Results

You might initially conclude that accounts receivable in 2025 has the potential for higher collectability risk simply because there were more customers with outstanding invoices in 2025 than in 2024. However, this initial judgment may be affected by internal assumptions and biases **(Risks)**:

- The 2025 increase in the number of customers with outstanding balances may not be a negative result, as the company grew and added new customers in 2025.
- These results may motivate you to evaluate alternative explanations and perform new analyses, such as determining if the 2025 customer numbers with outstanding invoices at year end had more invoices owed in 2025 than in 2024, or if the value of their outstanding balances were higher in 2025, possibly signaling a cash liquidity concern This result is more closely relevant to our company's accounts receivable risks of uncollectability **(Alternatives)**.

Diagnostic Analysis Strategies

Diagnostic analysis strategies can be compared to detective work. Accountants can use these analysis strategies to identify and discover the most likely causes of accounting phenomena. While both descriptive and diagnostic analysis strategies are descriptive techniques, diagnostic analysis strategies also identify what factors have caused the lower sales revenues or the higher costs, for example. Diagnostic strategies always require applying critical thinking skills.

Stakeholders rely on accountants across accounting practice areas to diagnose what is happening and explain, typically in non-accounting terms, why it is happening. For instance, systems accountants often use error dispersion data to identify system performance issues or unusual activity data that may indicate unauthorized party access.

Let's use the WeMakeIt Inc. accounts receivable data set again, this time to illustrate diagnostic analysis:

- **New objective question:** Do the returning credit customers have more outstanding invoices at the 2025 year-end than they had outstanding at the 2024 year-end?
- **Data strategy:** Once again, use the data set for the 2024 and 2025 year-end accounts receivable open invoices.
- **Analysis strategy:** Group the data by CustomerNumber for this question.

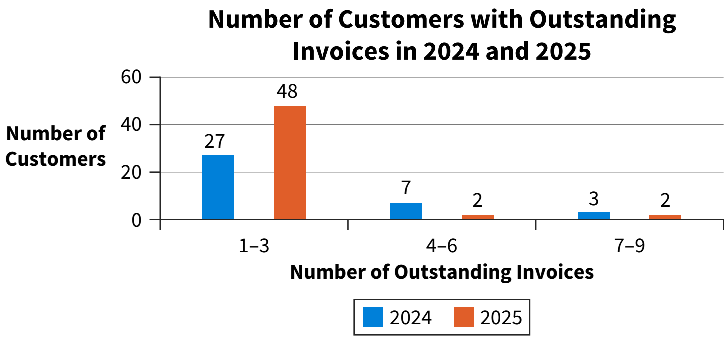
To answer this question, we may choose to create a new, calculated ordinal variable (recall that a **variable** is a data item that will be used in the analysis). This variable will group credit customers into increasing ranges based on the number of invoices they have outstanding at year-end. There are some benefits to this strategy:

- Changing a continuous variable into ordinal ranges results in meaningful groupings rather than a simple count of the number of invoices outstanding per customer.
- The new ordinal variable allows a better understanding of the overall results of the number of outstanding invoices in total.

How To

Let's create three levels for outstanding invoices: ranges of 1-3 invoices, 4-6 invoices, and 7-9 invoices as the new ordinal variable levels. **Illustration 4.16** provides a visualization of the results of this analysis strategy. (**Data** How To 4.2 describes how to recreate this visualization with a different tool.)

ILLUSTRATION 4.16 Frequency of Customers with Outstanding Balances



The results of this second analysis show that the number of customers with 4-6 and 7-9 outstanding invoices had decreased in 2025 compared to 2024. However, there are almost twice as many customers with 1-3 past due invoices in 2025 compared to 2024. These results are mixed news for concerns about future accounts receivable collectability risks.

This surprise finding is a good reminder about the importance of critical thinking and deliberation about the initial analysis results. Thinking critically when interpreting analysis results involves evaluating both the analysis strategies performed and not performed. Skeptically questioning data analysis strategies can provide valuable insights, as this final analysis did.

APPLYING CRITICAL THINKING 4.6: Controls for Risks to Analysis Strategies

Depending on the objective of the project, analysis strategies can involve a diverse mix of variables with different sources and measurement scales. Simply adding variables and complexities to analysis choices without justification can lead to information overload, which can impair your judgement. For example, sales forecasts might involve fewer variables in industries such as agriculture, as food demand is more stable, than for airlines, as travel is more discretionary. Professional risk-reducing habits include **(Risks)**:

- Keeping analysis as simple as possible by including the fewest yet most relevant variables.
- Documenting each step, especially the benefits and drawbacks of each choice as you develop, evaluate, and choose an analysis strategy **(Alternatives)**.

The process for analyzing WeMakeIt, Inc.'s accounts receivable, whether descriptive or diagnostic, involves an approach that can be used for most analysis strategies in accounting:

- Start with a clear objective and specific questions that should be answered.

- Develop data alternatives, evaluate each on factors related to the objective, and select the best data strategy. Filter the data set down to the variables and ranges of interest necessary for the objective.
- The best analysis strategies typically create new calculated within-row variables necessary for the analysis. These new calculated variables can be categorical groupings, ordinal rankings, interval measures, and ratio variables. Remember that relevant groupings can make it easier to interpret results.
- Calculate any vertical analysis measures necessary, such as totals, frequencies, averages, and measures of dispersion across the rows in the data set.
- Finally, the analysis variables can be correlated or sliced by dimension variables to add insights from disaggregating the vertical measure total.

Descriptive and diagnostic analyses help to better understand what happened and why it happened. What if the objective is to predict a future outcome or determine what strategies might achieve a specific outcome? In those cases, we need to use predictive or prescriptive analyses, which are explained next.

Designing Analyses to Predict and Prescribe

Predictive analysis strategies use historical data to create models that estimate a future value or outcome. Prescriptive analysis strategies, on the other hand, help determine which option is most likely to produce the best outcome given the objective. Predictive and prescriptive analysis strategies can determine the best use of resources, improve performance, anticipate market reactions and movements, and avoid process breakdowns:

- Accountants commonly use these analysis strategies to evaluate tax planning options, operational and marketing strategy options, financing options, investment options, and exit strategy options.
- Industries with these analysis objectives include lending institutions that want to avoid bad credit decisions and insurance companies that must anticipate natural disasters and disease onset so they can adjust business models before crises that cause their businesses to suffer happen.

Illustration 4.17 summarizes the most common predictive and prescriptive analysis strategies used by accountants that do not involve advanced technology tools. The green checkmarks indicate appropriate analyses, and the red x's indicate inappropriate analyses. As the illustration shows, there are not many predictive or prescriptive models for categorical and ordinal scale variables unless they are used in models in which interval or ratio scale variables are desired for the prediction.

ILLUSTRATION 4.17 Analyses Strategies for Predictive and Prescriptive Analytics

	Data Measurement Scale			
Potential Analyses	**Categorical**	**Ordinal**	**Interval**	**Ratio**
Trendlines	✗	✗	✓	✓
Regression analysis (linear)	✓	✗	✓	✓
Optimization models	✗	✗	✓	✓
What-if analyses	✗	✗	✓	✓

✗ = inappropriate analysis for the scale

Both predictive and prescriptive analysis strategies often use statistics and sophisticated modeling techniques for their algorithms:

- Common tools for this include statistical regression, time series analyses, process mining and mapping, text analysis, artificial intelligence models, data mining, mathematical modeling, complex simulations, and what-if sensitivity analysis modeling.
- These models train themselves by leveraging patterns, relationships, and structures within the existing data to accurately anticipate future conditions and outcomes.

The key to designing predictive or prescriptive strategies is to select data that will have the best potential to inform the predictive or prescriptive objective questions:

- If the goal is to predict next year's production equipment maintenance expense, data that helps explain the expected amount or increases or decreases in maintenance expenses are necessary. In other words, what variables drive changes in maintenance expense? Once those variables are identified, the related data can be selected and prepared for analysis.
- If we believe production level impacts maintenance expense, then include a data variable for the number of products produced, a measure of the most common levels of production, or a measure of whether production is expected to increase or decrease from current levels.

Suppose the WeMakeIt's auditors are testing the assumptions used to create the balance in the the company's allowance for doubtful accounts. The auditors developed a model that included three variables that might impact whether the invoices would be collected on time. The model predicts how many days outstanding an invoice may be based on the invoice amount, if they are a new (1) or existing customer (0), and the credit period allowed on their invoice. The result of the regression model is provided in **Illustration 4.18**.

ILLUSTRATION 4.18 Prediction Model for Accounts Receivable Days Outstanding

Regression Statistics					
Multiple R	0.825790307				
R Square	0.681929631				
Adjusted R Square	0.681542056				
Standard Error	3.549920129				
Observations	2466				
ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	66518.36434	22172.78811	1759.475173	0.0000
Residual	2462	31025.95885	12.60193292		
Total	2465	97544.3232			
	Coefficients	*Standard Error*	*t Stat*	*P-value*	
Intercept	−7.961650225	0.670515711	−11.87392047	1.16228E-31	
InvoiceAmount	0.000829906	0.000903172	0.918879698	0.358248513	
New or Existing Customer	1.535043063	0.189912705	8.082887691	9.81036E-16	
Credit Period in Days	0.395420024	0.006452145	61.28504622	0.000	

This model reveals that these variables (invoice amount, new or existing customer, and credit period) explain about 68% of the change in days outstanding for a customer (adjusted *R*-square). The model is significant with a Significance *F* of less than 0.05. The negative intercept corrects the model's predictions by almost 8 days because customers rarely pay within the first week when they are offered an interest free period of time to pay their invoices. The significance of the intercept indicates that the mere existence of a credit period is a variable that impacts the number of days customers take to pay their invoices. With this model, the auditors can now predict that a new customer with a $900 invoice and 60 days to settle will take about 16.5 days to make their payment:

$$-7.9617 + \$900\ (0.0008) + 1(1.5350) + 60(0.3954) = 16.48 \text{ days}$$

Analysis Strategy Risks and Suggested Controls

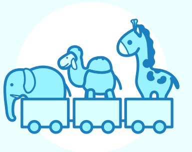
The last two steps of a data analysis plan are to consider risks inherent to the data and analysis strategies and to put in place controls to reduce those risks. **Illustration 4.19** captures the most common risks that accountants face when analyzing data.

ILLUSTRATION 4.19 Common Analysis Risks and Recommended Controls

	Risk	Control
Analysis	The analysis is not appropriate for the variable's measurement scales.	Review whether the analysis is appropriate for variables' measure scales. Change analysis strategy if needed.
	The sample size is too small to allow for accurate inferences.	Gather more data to meet the model's minimum data points and assumptions, or select data analyses that do not require large samples.
	The sample is not representative.	Test the reasonableness of sample with descriptive analyses.
	Confounding variables are interfering with the reliability of the information models.	Perform the analysis with and without the confounding variable to test for explanatory power differences.
	The omission of relevant variables, or variables that are unknown and not controlled for, resulting in less reliable modeling.	Add available proxies for the missing variables to see if the model's explanatory power increases.
	Uncorrected and uncontrolled dirty data issues, such as errors, empty fields, outlier data points, and data from different levels of quality sources.	Properly prepare the data before starting the analysis. (Data preparation is explained in the next chapter.)
	Insufficient knowledge about or experience in performing the analysis strategy or interpreting the results.	Acquire proper training or work with a team that has more collective knowledge.
	Poorly made or misleading visualizations.	Perform descriptive analyses to compare and augment the visualization. Use visualization best practices.
	Failing to address the underlying uncertainties in the context.	Perform sensitivity analysis on all assumptions used in the analysis decisions.
	Insufficiently trained prediction and classification models.	Include more data to train the models and fully verify their accuracy before using them on the test data.

It is critical to consider these risks during analysis strategy planning and to add controls to help mitigate them. Without those controls we risk preparing, interpreting, and reporting incorrect analyses results, possibly causing ourselves or our stakeholders to make harmful decisions. Now that we have worked through the process of choosing the best data and analysis strategies, let's examine how accountants do this in professional practice.

Apply It 4.3
Create a Predictive Data Analysis Project Plan



Financial Accounting Tremendous Toys has implemented a database to capture inventoriable costs from their production each week. Data from the following data fields have been extracted, noting each field's data type and measurement scale.

Field Name	Field Type	Field Measurement Scale
ProductionRunNumber	Non-measured raw data	Categorical (Nominal)
ProductionRunDate	Non-measured raw data	Interval
NumberOfToysinRun	Measured raw data	Ratio
ToyIDNumber	Non-measured raw data	Categorical (Nominal)
DirectMaterialsUsed	Measured raw data	Ratio
DirectLaborUsed	Measured raw data	Ratio
OverheadCostApplied	Calculated data	Ratio
TotalRunCost	Calculated data	Ratio
TotalUnitCost	Calculated data	Ratio

Using this production data, as well as the sales data for how many toys of each type were sold each month for the last three years, create a predictive data analysis project plan that will estimate Tremendous Toy's year-end inventory valuation at the end of the next fiscal year. Your project plan should address the following:

1. Project objective and specific questions
2. Data strategy
3. Analysis strategy
4. Risks in data and analysis strategies
5. Controls for data and analysis strategies

SOLUTION

1. Project objective Specific questions	• Accurately predict the next year-end's inventory value. • How many units are predicted to be in inventory at the next year end? What is the predicted unit cost for each toy type at the end of next year?
2. Data strategy	• Data from the last three years of Tremendous Toys' production records will be extracted and prepared for analysis regarding materials costs, labor costs, and overhead charges. • The sales units data by toy type for the last three years will be extracted, cleaned for missing, duplicate, and outlier data errors, and aggregated by month. This results in 36 data points for the monthly unit sales of each toy type. • A calculated variable for average monthly cost for each toy type will need to be created using the NumberofToysinRun and TotalRunCost fields. First, this data must be aggregated by month. Then, for each toy in each month, the total monthly production cost should be divided by the total number of toys produced to calculate the average toy cost per month for each toy type. The result will be 36 data points for the monthly average cost for each type of toy.
3. Analysis strategy	• The December sales in units can be estimated by running a regression on each month's unit sales. • A calculated variable must be created for each month's ending inventory units for each toy type, resulting in 36 data points for each toy. This variable's formula is: beginning inventory units + monthly toys produced – monthly units sold. (You can assume that this business started three years ago with no beginning inventory.) • The December year-end ending inventory in units can be estimated by running a regression on the monthly ending inventory units for each toy type. • The December year-end average cost per toy can be estimated by running a regression on the average monthly costs per toy. • To calculate the total December ending inventory valuation, the December ending inventory units for each toy should be multiplied by its corresponding estimated December average cost. Then these total costs per toy must be added for the total December year end inventory value.

(Continued)

4. Risks to data strategy	• There may be missing, duplicated, or erroneous data in the fields selected for analysis.
Risks to analysis strategy	• Costs may have changed over the three years due to seasonality, changes in vendors, and inflation. • Calculations may contain errors.
5. Data strategy controls	• Perform data cleaning tests to find and correct missing data, duplicates, and errors that result in outlier data points.
Analysis strategy controls	• Compare the unit cost and sales per month to determine if an inflation factor, seasonality, or other changes have occurred. If data shifts are noticed, their cause and permanence should be verified by speaking with the employees who would know the answers. • Each of the calculations performed should be verified, as well as the averaging calculations. • Regression analyses can be verified by making visualizations of the three years of monthly data points and visually inspected for agreement with the regression estimations.

4.4 How Do Data and Analysis Strategies Differ Across Practice Areas?

LEARNING OBJECTIVE 4

Summarize data and analyses strategies in professional practice areas.

As the last step of the data analysis planning stage, accounting professionals across practice areas design data and analysis strategies based on their project objectives. Here, we offer examples of common data and analytics strategies for the accounting information systems (AIS), auditing, financial accounting, managerial accounting, and tax accounting professional practice areas.

Accounting Information Systems

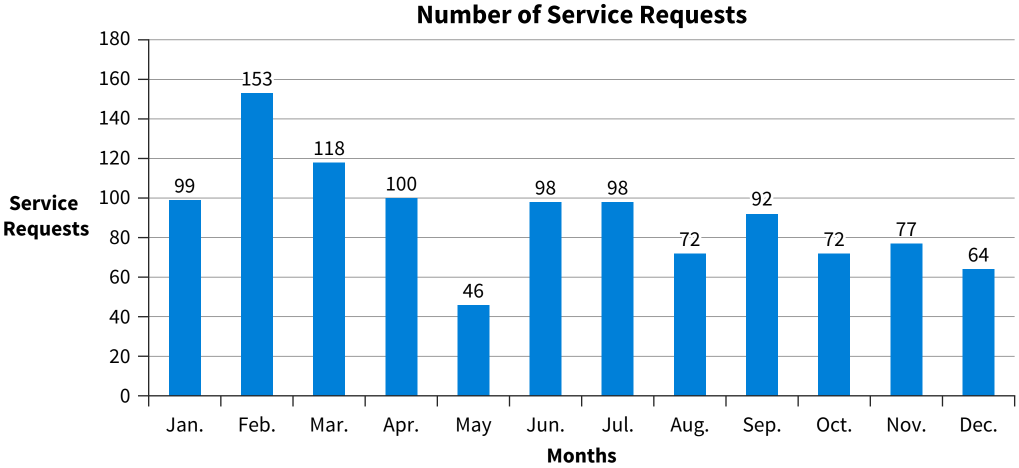
Because a company's accounting information system is involved in planning, executing, controlling, and reporting the business' operations, data analysis projects in this area often involve interdisciplinary data and knowledge. These project objectives can range in scope from impacting just one or two employees to involving most of the organization. All four types of analytics objectives (descriptive, diagnostic, predictive, and prescriptive) are commonly used.

The data analyzed in these projects can include a variety of IT operational data with different levels of data integrity and control documentation. More traditional accounting data are often analyzed with qualitative categorical and ordinal data:

- Counts of help tickets and incidents.
- Error counts and delay times.
- Login issues.
- System satisfaction and IT service satisfaction.

For example, **Illustration 4.20** is an example of a visualization of the number of monthly service requests related to the accounting information system of a company. The objective was to identify how many and which months experienced more than the IT staff's capacity of 90 service requests.

ILLUSTRATION 4.20 Number of Service Requests Each Month



AIS data analyses projects may also choose data strategies which use quantitative interval and ratio data:

- Costs such as equipment and software installations and updates, and IT staff labor.
- Budget variances for any of these costs.

It is important to select data that captures the AIS system's performance, vulnerabilities, and errors. For example, to analyze performance we could analyze the number of spam emails (ratio data) before and after investment in a new firewall.

Accountants designing analysis strategies for AIS projects are focused on increasing competitive advantage and improving the organization's operations. They often use statistics to understand which element of the accounting system they should focus on. The mathematical formality of statistical tools can help to persuade management to make the necessary investments.

There are risks involved with AIS data choice and analysis choices. Some common strategic and critical risks and suggested controls for AIS data and analysis are listed in **Illustration 4.21**.

ILLUSTRATION 4.21 Risks and Controls for AIS Data and Analysis Strategies

	Risks	Controls
Data	• Human behavior is not captured. • Data manipulation from unauthorized access or untrained employees. • Loss of data due to security breaches, ransomware, or natural disasters. • Difficulties capturing high velocity data. • Difficulties measuring interdepartmental dependencies on process software. • Dirty data and outliers.	• Strong logs and segregation of duties in IT. • Strong access controls. • Effective back ups and graceful degradation. • Clear procedures. • Thorough data cleaning and triangulation to verify data capture. • Strong audit trails. • Data and change management controls. • Data retention and destruction policies.
Analysis	• Errors in variable transformations. • Errors in calculations. • Data availability windows are too short to capture more strategic trends and issues.	• Verifying analysis steps and calculations. • Performing sensitivity analyses for critical results. • Creating effective visualizations and explanations.

Auditing

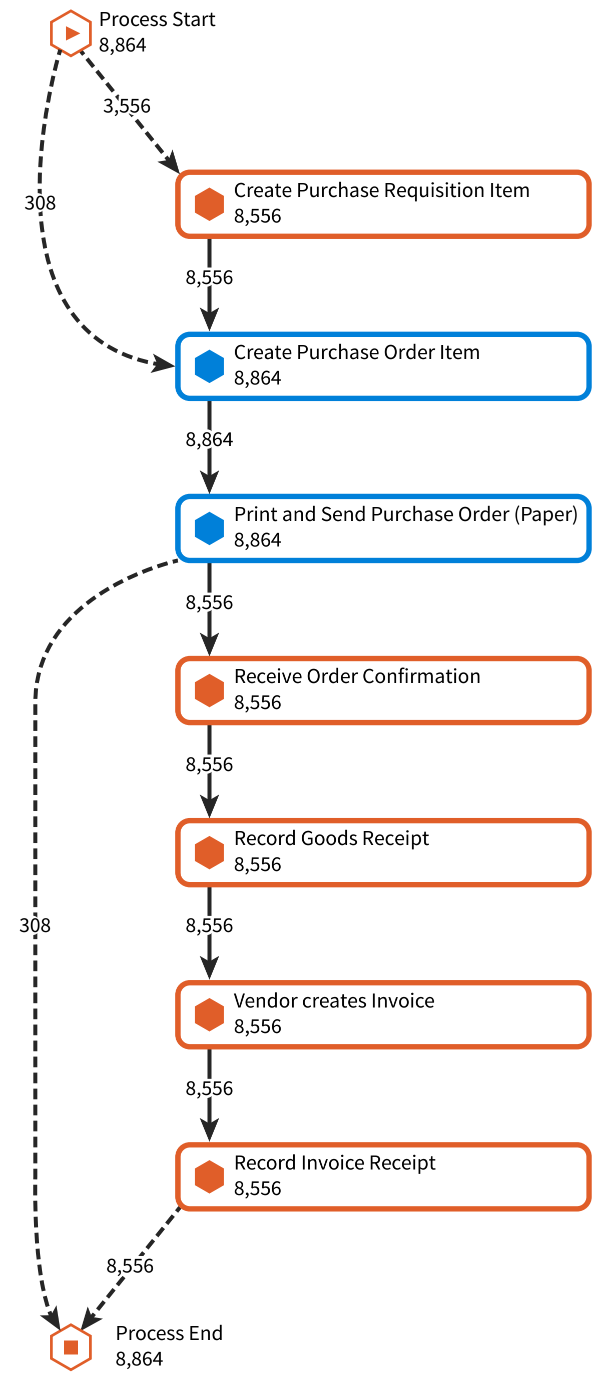
Auditors' historic resistance to change, due to high litigation risks and Public Accounting Oversight Board (PCAOB) review of audit processes, is disappearing as new technologies help them perform higher-quality, lower-risk, and more efficient audits. Audit firms are increasingly investing in technology and new ways to use it. They are reorganizing to offer new services to provide more business intelligence and economic value to their clients.

Auditors use statistics to determine risks to account balances and unusual transactions, especially in journal entries that involve estimates and assumptions. They commonly work with different kinds of data sources, from process documentation, journal entries, general and subsidiary ledgers, trial balances, to financial ratios. Auditors use analysis strategies for:

- Continuous auditing modules to test large data populations rather than testing samples with inference risks.
- Automated identification of dirty data, unusual transactions, and pattern anomalies to better reduce audit risk and fraud risks.
- Testing entire transaction cycles' internal controls with process mining and tracing purchasing to payment flows, P-cards (purchasing credit cards) usage and payroll documentation. Testing the revenue cycle from order to collection.
- Using robotic process automation (RPA) to remove the human (and often inconsistent) element of repetitive audit tasks, freeing auditors to focus on areas that require thoughtful critical thinking and judgments. (RPA applications and benefits to accountants are explained in the chapter that covers data and analysis developments in accounting.)
- Testing hypotheses about inventory and fixed assets with sampling, statistical tests, and inferences for the population of these assets.

Illustration 4.22 is an example of auditors using a process mining analysis strategy to learn whether all purchases followed the correct process from requisition to the recording of the invoice for the purchase. The visualization indicates that more than expected purchases (308 purchases) skip the authorized purchase requisition process step as well as the authorized order confirmation, goods receipt, and vendor invoice steps. These deviations from the authorized process are concerning to an auditor. The blue boxes indicate where the process is letting in purchase orders without purchase requisitions and letting out purchase orders without the subsequent required steps. An auditor will then want to follow-up on these exceptions to the authorized process.

ILLUSTRATION 4.22 Process Mining Results for Purchases Analysis



Regardless of the data and analysis strategies used, auditors must identify the risks and controls necessary to assure they can rely on the results of their analyses (**Illustration 4.23**).

ILLUSTRATION 4.23 Risks and Controls for Auditing Data and Analysis Strategies

	Risks	Controls
Data	• Missing data. • Dirty data. • Fake transaction data.	• Strong audit trail tests of internal controls for data entry, storage, processing, and reporting. • Testing authorization and access controls.
Analysis	• Errors in assumptions. • Errors in calculations and conclusions.	• Verification of all audit steps and calculations. • Checks on authentication and approvals. • Sensitivity analysis for critical results.

Financial Accounting

Financial accountants are primarily responsible for capturing, recording, processing, storing, and reporting accounting information. Accuracy, thoroughness, and documentation are essential. The accounting data they use for analyses are both guided and restrained by accounting regulations and government agency compliance. Due to the measurement scales of the variables they need for analysis, financial accountants may have to transform data depending on their objective questions. These data most often include categorical, interval, and ratio data.

The purpose of a data analysis strategy might be describing and diagnosing based on corresponding financial accounting rules and regulations:

- Nature, timing, and authorization of transactions (and diagnose issues) charged to each account.
- Internal control effectiveness and weaknesses.
- The completeness and reasonableness of adjusting entries at the end of the period.

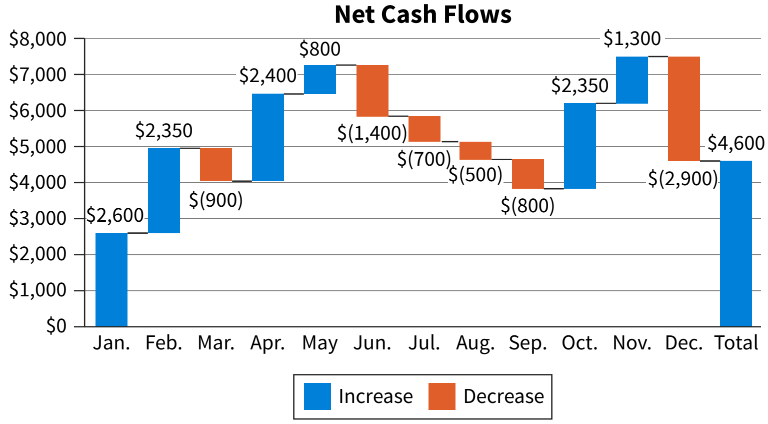
Financial accountants also use analysis strategies designed for their predictive and prescriptive objectives regarding financial outcomes for managers and board members. One example is estimating additional information to be presented along with their financial statements, such as:

- Ranking alternative capital sources and costs of capital in terms of favorability.
- Future net income and cash flows from operations, investing, and financing activities.
- Impacts of new strategies on the financial statement.
- Expected future costs associated with contingent liabilities, pension costs, new business units or the discontinuation of business units.

Financial accountants use statistics to identify opportunities and problems with profitability, liquidity, and business valuation. They analyze categorical industry codes from their client's categorical groups of held to maturity, trading, and available for sale investment assets to ensure desired diversification risks and holding strategies.

Illustration 4.24 is a waterfall chart that shows the analysis results for a project with the objective to better understand the monthly net cash flows for an organization by month. A waterfall chart is a useful visualization to show the positive and negative components of a change.

ILLUSTRATION 4.24 Net Cash Flow Visualization



This waterfall visualization shows the change in cashflow for each month. For instance, in February cashflow increased by $2,350 but then decreased by $900 in March. The net of all the monthly changes agrees to the total. Like other accountants, financial accountants must identify potential risks in their data and analysis strategies (**Illustration 4.25**).

ILLUSTRATION 4.25 Risks and Controls for Financial Accounting Data and Analysis Strategies

	Risks	Controls
Data	• Missing operational and adjusting entries. • Dirty data. • Fake transaction data. • Incorrect valuation. • Incorrect account. • Incorrect time period. • Missing data.	• Strong accounting cycle steps. • Strong internal controls for data entry, storage, processing, and reporting. • Authorization and access controls. • Testing for and correcting missing, duplicate, and outlier data.
Analysis	• Calculation errors. • Biases and self-interests. • Incorrect calculations. • Not following GAAP analysis guidance.	• Verifying all cycle steps and calculations. • Checking authentication and approvals. • Sensitivity analysis for critical results. • Verify applicable GAAP guidance.

Managerial Accounting

Managerial accountants add value to their organization with different data analysis strategies. The purpose of most managerial accounting data analyses is to improve planning, operational control, and decision-making to support an organization's mission and strategies. This type of data analysis is valuable for improving the selection, execution, and evaluation of organizational strategy. These kinds of analyses can also lead to decisions that give employees more access to information, which improves performance and, eventually, organizational culture and operations.

APPLYING CRITICAL THINKING 4.7: Use Multiple Disciplines in Managerial Accounting

The knowledge base needed by managerial accountants is broad, involving multiple disciplines. For example, to reduce factory operating costs, managerial accountants must understand the engineering of their production machines and know how to create reliable statistical models for future production **(Knowledge)**:

- Accounting and economic concepts
- Organizational behavior
- Operations management
- Technology tools
- Quantitative analyses disciplines

Managerial accountants prepare analysis strategies, both for routine and ad hoc (one-time) objectives for each functional area of their organization. These strategies use data across measurement scales to describe, diagnose, predict, and prescribe strategy impacts:

- Identifying areas where innovation in business organization, policy and process will increase efficiencies, for example, to reduce non-value-added steps and delays.
- Identifying areas where innovation in business partnerships, operations, and internal controls will increase effectiveness.
- Increasing competitive advantage through many new business intelligence opportunities.
- Improving compliance with all legal and regulatory regulations.

Managerial accountants may perform data analysis to determine the potential business process and internal control benefits of automating the purchase requisition to vendor invoice process (see Illustration 4.22). Automation could force all departments to go through the authorized business process rather than going around it. Identifying risks to the data and analyses used in managerial accounting is important to ensure analysis results are accurate and reliable (**Illustration 4.26**).

ILLUSTRATION 4.26 Risks and Controls for Managerial Accounting Data and Analysis Strategies

	Risks	Control Choices
Data	• Dirty data. • Missing data. • Irrelevant data.	• Independent verification. • Segregation of duties. • Sound personnel governance. • Verify authorization. • Considering the purpose when making data choices.
Analysis	• Errors in calculations. • Errors in attributions. • Biases and self-interests.	• Verifying all steps and calculations.

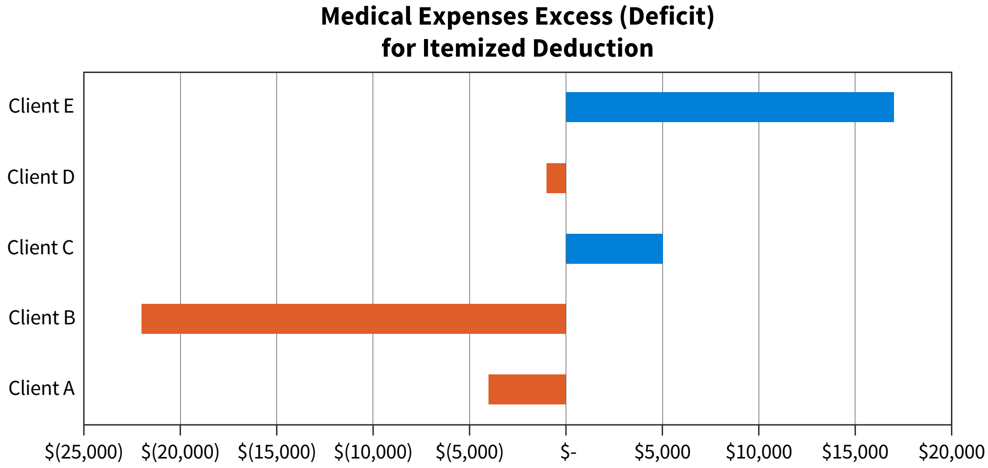
Tax Accounting

Thanks to data analytics, tax accountants have more information for their tax planning and compliance services, which improves their judgments and how they document defend their positions to both clients and regulators. Recent surveys show that over 58% of tax accountants are using data and analytics strategies for many areas of their professional practice[1], including:

- Sales and use taxes and local taxes (SALT).
- Value added taxes (VAT).
- Goods and services taxes (GST).
- Customs duties.
- Transfer pricing and intercompany transactions.
- Tax compliance.
- Tax provisions.

Perhaps the biggest effect of data analysis strategies on tax practice is the movement away from a dependency on historic data and toward a forward-looking, value-adding service perspective by using predicting and prescribing analytics strategies. Examples of these predictive and prescriptive analytics include using complex data modeling to evaluate decision and position alternatives, which improves the value and accuracy of advice. **Illustration 4.27** is a live dashboard that shows current year medical expenses compared to the level needed for itemized deductions for five of their individual clients. This would help the tax accountant provide advice to these clients.

ILLUSTRATION 4.27 Medical Expenses Itemized Deduction Threshold Assessment by Client



Another example is sophisticated querying of clients' accounting data to better identify opportunities, such as when to buy or sell assets, and understanding the operational decisions driving tax liabilities.

APPLYING CRITICAL THINKING 4.8: Think Critically in Tax Accounting

- Tax accountants consider their clients, tax authorities, and their tax firm partners when they plan different data analyses. Each of these stakeholders may have different objectives and priorities. For example, clients may prioritize minimizing their tax obligations, so that would be the objective of your tax planning services **(Stakeholders)**.
- Strong research and analysis skills are necessary, such as when performing tax research, developing tax strategies, and estimating tax liabilities. For example, state laws are different than the national and international law for every type of tax category **(Knowledge)**.
- Clients, both individuals and organizations, may have incentives to omit taxable revenues and include irrelevant and inappropriate expenses. These data risks should be controlled, perhaps with interview questions as well as verifying the records provided through indirect tests of reasonableness **(Risks)**.

[1]International Tax Review (2019). Preparing for the future of tax. https://www.internationaltaxreview.com/article/2a6a41udxe4e2kuc8zoqo/preparing-for-the-future-of-tax (accessed 13 July 2022).

Identifying risks and assuring that proper controls are in place to mitigate risks are also critical (**Illustration 4.28**).

ILLUSTRATION 4.28 Risks and Controls for Tax Accounting Data and Analysis Strategies

	Risks	Control Choices
Data	• Dirty data and outliers. • Missing data.	• Implement backups. • Strong audit trails. • Tax management controls. • Data retention and destruction policies. • Data authenticity and dirty data verification and correction.
Analysis	• Errors in data evaluations and tests. • Errors in calculations and forms. • Lack of tax code knowledge.	• Verification of all tax rules' liability steps and calculations. • Sensitivity analysis for critical results. • Effective visualizations and explanations.

Apply It 4.4
Match Strategies with Professional Practice Areas

For each of the following data and analyses objectives:

1. Identify the data measurement scale that would be utilized.
2. Identify one practice area that is likely to perform that analysis.

Some of the options have more than one correct answer for both (1) and (2), but only enter one correct answer for each.

Data Measurement Scales: categorical, ordinal, interval, ratio

Practice Areas: Accounting information systems, Auditing, Financial accounting, Managerial accounting, Tax accounting

Objective	Measurement Scale	Practice Area
Aging of accounts receivable's year-end open invoices on days outstanding.		
Predicting cost savings and productivity increases from new AIS investments.		
Creating dollar value stratified sampling plans for investment asset accounts.		
Evaluating web server security incidents by type of incident.		
Prescribing product revenue increases by product from marketing campaigns.		
Predicting future cash flows from operations.		
Calculating before and after process improvements from system investments.		
Testing different tax planning strategies by evaluating future cash savings.		
Estimating inventory counts at year-end by sampling a few of the stores and making population inferences.		
Testing approvals and documentation for transaction authorization.		

SOLUTION

Objective	Measurement Scale	Practice Area
Aging of accounts receivable's year-end open invoices on days outstanding.	Interval	Financial and managerial accounting
Predicting cost savings and productivity increases from new AIS investments.	Ratio	Managerial accounting
Creating dollar value stratified sampling plans for investment asset accounts.	Interval	Auditing
Evaluating web server security incidents by type of incident.	Categorical	Accounting information systems
Prescribing product revenue increases by product from marketing campaigns.	Ratio	Managerial accounting
Predicting future cash flows from operations.	Ratio	Managerial accounting
Calculating before and after process improvements from system investments.	Interval	Accounting Information systems
Testing different tax planning strategies by evaluating future cash savings.	Ratio	Tax accounting
Estimating inventory counts at year-end by sampling a few of the stores and making population inferences.	Ratio	Auditing
Testing approvals and documentation for transaction authorization.	Categorical	Financial accounting

Chapter Review and Practice

Learning Objectives Review

1 Identify the components of a data analysis project plan.

A data analysis project plan is a blueprint to follow for the project. Using a planning tool can help choose the best data and analysis strategies for the project objectives. There are five steps:

- Step 1: Focus on the project's objective and the specific questions that must be answered.
- Step 2: Decide which data are needed to answer the questions, develop alternatives, evaluate, and rank the best option.

- Step 3: Consider the methods available to do the analysis and rank the alternatives. Evaluate and choose the best strategy option.
- Steps 4 and 5: Consider critical risks inherent to both the data strategy and analysis strategy.
- Steps 4 and 5: Embed controls to reduce these risks in data strategy and analysis strategy.

2 Describe how to develop a data strategy.

A successful project hinges on using the appropriate data and evaluating its characteristics. Considering the data's potential impact on the analysis and any risks makes it easier to select the most suitable data for a project's goals:

- The first step is selecting the most appropriate data. Appropriate data are data that are available and relevant to the data analysis objective questions.
- The data's measurement scales must also be suitable for use with the chosen data analysis method, as not all analyses are valid for every measure scale. Data can have categorical, interval, ordinal, or ratio scales.
- Data can be either internal (generated within the organization), external (obtained from a source outside of the organization), or a combination of both. Data can be raw measures or assigned non-measures, and can also be calculated fields.
- Consider potential risks related to the data chosen for the analysis and put controls in place to mitigate those risks. Three common data risks (and controls) include: non-representative samples (verify representativeness of the sample), outlier data points (identify outliers and then justify or remove), and dirty data (verify integrity of the data set and clean).

3 Explain how an analysis strategy is designed.

There are two things to consider when designing a data analysis strategy: (1) the objective questions and (2) the measurement scale of the data. There are four data analysis objectives: descriptive, diagnostic, predictive, or prescriptive. The questions we design are based on the objective:

- When designing any type of analysis, the appropriate analysis is dependent on the measurement scale of the data. Typical descriptive and diagnostic analysis methods include measure of central tendency (mean, median), frequency distributions, measures of data dispersion (e.g. range, quartiles, standard deviation), visualizations, correlation, and calculations (e.g. totals, percent change).
- Typical analyses used for predictive and prescriptive analyses include trendlines, regression analysis, optimization models, and what-if analyses.
- There are many common risks inherent in descriptive, diagnostic, predictive, and prescriptive analyses that can be identified and controlled. Some of these risks include performing an analysis with dirty data, or analysis not appropriate for the variable's measure scales, using too small a sample, or a non-representative sample, or including (or omitting) variables from the analysis.

4 Summarize data and analyses strategies in professional practice areas.

Each professional accounting practice area plans different data and analyses strategies for varying objectives, using data with a range of measurement scales to describe, diagnose, predict, and prescribe results for their stakeholders and objectives. The risks inherent to their data and analysis choices, as well as the controls they implement are a function of their roles, responsibilities, and project objectives:

- Accounting information systems professionals plan data and analysis strategies for AIS governance, strategic planning and evaluation, operational performance, and compliance objectives. Their projects typically involve business process functionality, business intelligence, and cybersecurity purposes.
- Auditors plan data and analyses that have professional audit process compliance and audit quality objectives to verify the representativeness of their clients' financial statement balances or their organization's accounting systems. External auditors have the additional data risks of using data that has been captured and extracted by their clients, oftentimes not able to verify the data completeness or accuracy.
- Financial accountants focus on regulatory compliance regarding transaction capturing, recording, processing, storing, and reporting objectives.
- Managerial accountants have a wide variety of internal governance, strategic, financing, investing, and operational objectives for their data and analysis plans.
- Tax accountants focus on both compliance and more forward-facing tax planning opportunities to serve their clients and organizations with their data analysis strategy plans. Similar to external auditors, tax accountants often must use data sets prepared by their clients, which may have greater risks of inaccuracy and incompleteness.

Key Terms Review

Attributes 4-12
Calculated data 4-14
Data set 4-12
Dirty data 4-16
Fields 4-12
Measured raw numeric data 4-13
Measurement scale 4-14
Non-measured raw data 4-13
Records 4-12
Variable 4-20

How To Walk-Throughs

How To

HOW TO 4.1 Calculate Year-End Bad Debts Estimation Using Excel

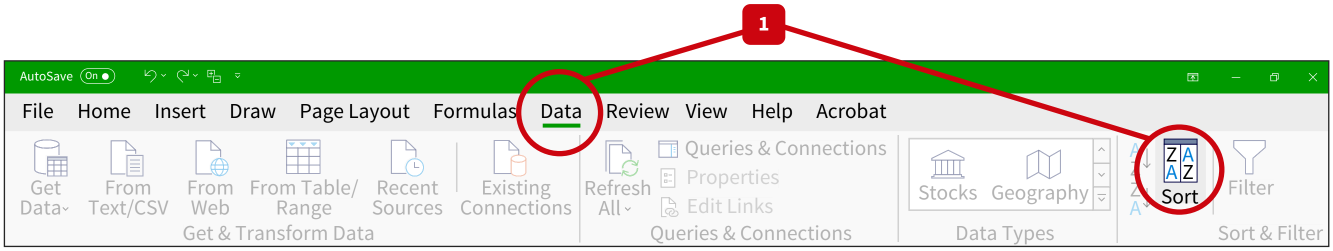
The figures shown in the table in Illustration 4.2 were calculated in Excel. Recall a few facts about this example:

- WeMakeIt, Inc. estimates that 2% of its 30 to 60 days outstanding receivables will be uncollectible, and that 30% of outstanding receivables older than 60 days will be uncollectible.
- The 2025 year-end allowance for uncollectible accounts unadjusted balance is $3,000.

What You Need: **Data** The How To 4.1 data file.

STEP 1: Select the entire data set, but not the labels, and sort the outstanding invoice data by invoice date. The **Sort** option appears in the **Data** tab. (**Illustration 4.29**)

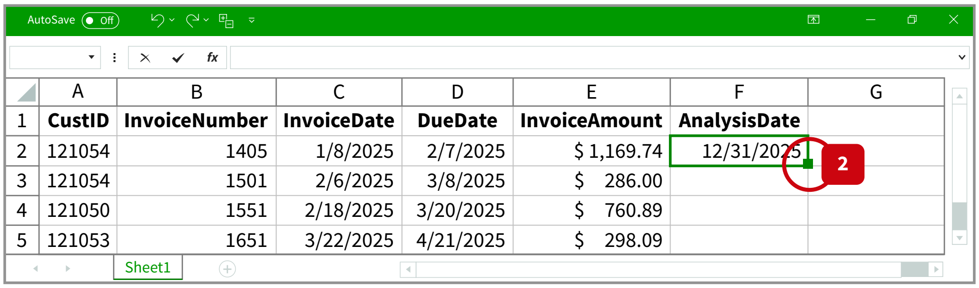
ILLUSTRATION 4.29 Sort Outstanding Invoices by Date



STEP 2: Create a new column to the right of the data for the year-end date of 12/31/2025:

- Add a label name for this column in row 1: "Analysis Date."
- In row 2, enter "12/31/2025" and copy this date to the bottom row of data. This can be easily done by clicking on the bottom corner of the cell where "12/31/2025" was entered and dragging the cursor down to the last row of data (**Illustration 4.30**).

ILLUSTRATION 4.30 Add Analysis Date Column

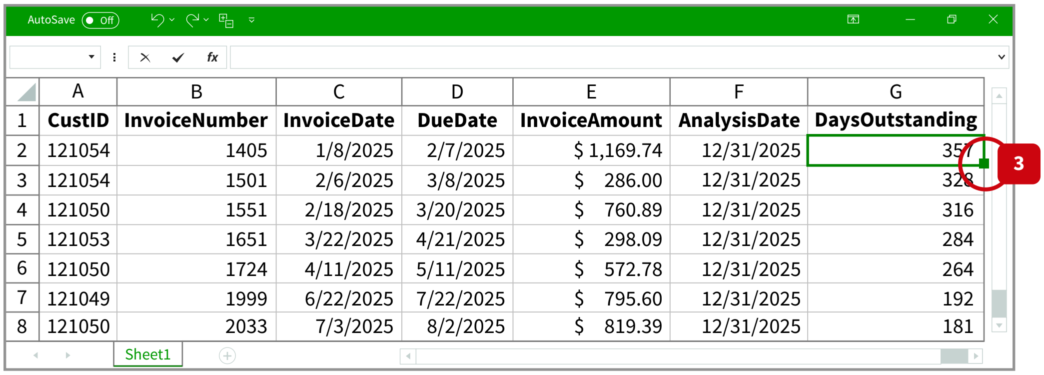


	A	B	C	D	E	F	G
1	CustID	InvoiceNumber	InvoiceDate	DueDate	InvoiceAmount	AnalysisDate	
2	121054	1405	1/8/2025	2/7/2025	$ 1,169.74	12/31/2025	
3	121054	1501	2/6/2025	3/8/2025	$ 286.00		
4	121050	1551	2/18/2025	3/20/2025	$ 760.89		
5	121053	1651	3/22/2025	4/21/2025	$ 298.09		

STEP 3: Create a new column to the right of the 12/31/2025 date column:

- Add a label name for this column in row 1: "Days Outstanding."
- Calculate the days outstanding by creating a formula in cell G2 which subtracts the invoice date (Cell C2) from 12/31/2025 cell (F2) for the Analysis Date. The formula is =F2-C2.
- Copy this formula through all the data rows in Column G. Copying can be done by clicking on the bottom corner of the cell where the formula was entered and dragging in down to the last row of data, (**Illustration 4.31**).

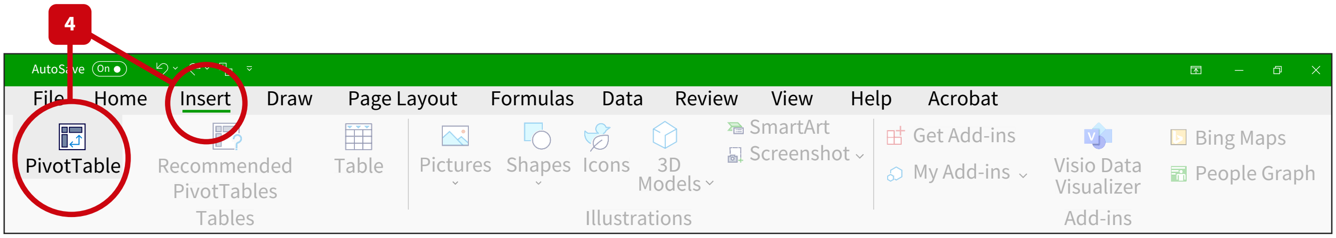
ILLUSTRATION 4.31 Data set with Analysis Date and Days Outstanding Columns



	A	B	C	D	E	F	G
1	CustID	InvoiceNumber	InvoiceDate	DueDate	InvoiceAmount	AnalysisDate	DaysOutstanding
2	121054	1405	1/8/2025	2/7/2025	$ 1,169.74	12/31/2025	357
3	121054	1501	2/6/2025	3/8/2025	$ 286.00	12/31/2025	328
4	121050	1551	2/18/2025	3/20/2025	$ 760.89	12/31/2025	316
5	121053	1651	3/22/2025	4/21/2025	$ 298.09	12/31/2025	284
6	121050	1724	4/11/2025	5/11/2025	$ 572.78	12/31/2025	264
7	121049	1999	6/22/2025	7/22/2025	$ 795.60	12/31/2025	192
8	121050	2033	7/3/2025	8/2/2025	$ 819.39	12/31/2025	181

STEP 4: Select the data field titles and data rows and create a PivotTable. This option is located under the **Insert** menu option on Excel (**Illustration 4.32**).

ILLUSTRATION 4.32 Insert PivotTable

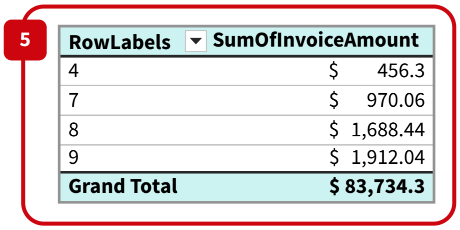


Select **New Worksheet**. Your screen will automatically move to the new worksheet. Name this table "BadDebts2025." The name appears at the top left of your screen.

STEP 5: Select any place in the PivotTable so the **PivotTable Fields** appears on the right of your screen:

- Drag **Invoice Amount** to the **Values** area.
- Drag **Days Outstanding** to **Rows**. The new PivotTable solution should appear (**Illustration 4.33**).

ILLUSTRATION 4.33 Unpaid Invoices by Days Outstanding

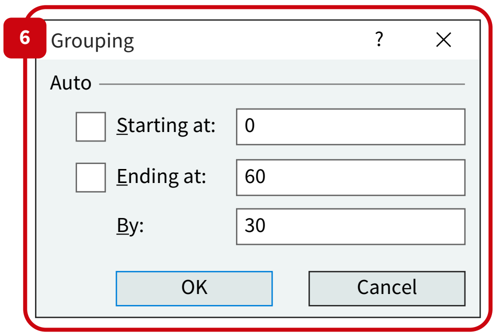


RowLabels	SumOfInvoiceAmount
4	$ 456.3
7	$ 970.06
8	$ 1,688.44
9	$ 1,912.04
Grand Total	**$ 83,734.3**

STEP 6: Next, select any row in the **Days Outstanding** column:

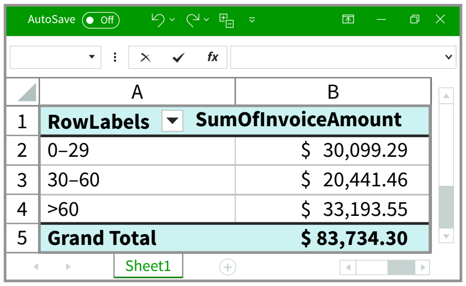
- Right click and select **Group**.
- In the dialog box that opens (**Illustration 4.34**), enter a zero (0) in the starting point, 60 in the ending point, and then enter 30 for the **By** option. Select **OK**.

ILLUSTRATION 4.34 Dialog Box to Group by Days Outstanding



Now the PivotTable has become three rows for the three desired age categories, with the subtotals already calculated for the **Sum of InvoiceAmount** column. Format the **Sum column** as currency (**Illustration 4.35**). Format the amounts as currency.

ILLUSTRATION 4.35 Age Group Totals for Accounts Receivable Invoice Values



	A	B
1	RowLabels	SumOfInvoiceAmount
2	0–29	$ 30,099.29
3	30–60	$ 20,441.46
4	>60	$ 33,193.55
5	Grand Total	$ 83,734.30

STEP 7: In the cell to the right of the SumofInvoiceAmount title, enter "Estimation %." In the cell to the right, enter "$ Uncollectible." Adjust column widths:

- In the cell to the right of the $ 20,441.46, enter "0.02," and in the cell below enter "0.3" for the uncollectible percentages.
- Format the **$ Uncollectible** column as currency.

STEP 8: In the cells to the right of each of these percentages, enter the following formulas using cell references:

- For the 30-60 day group, the formula is:

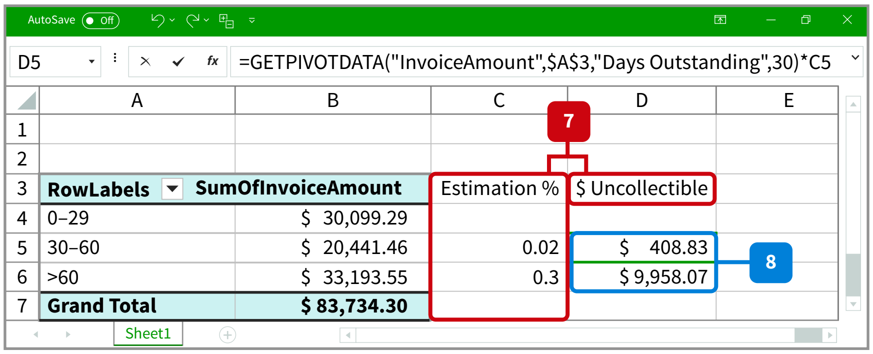
  =(click on cell with 20441.46)*(click on the cell with the 0.02)

- And for the >60 days group:

  =(click on the cell with 33193.55)*(click on cell with the 0.3)

- An example of the formula and totals that result is shown in **Illustration 4.36**. Add an underline to the cell containing the last formula.

ILLUSTRATION 4.36 Age Group Totals for Bad Debt Estimations



	A	B	C	D	E
1					
2					
3	RowLabels	SumOfInvoiceAmount	Estimation %	$ Uncollectible	
4	0–29	$ 30,099.29			
5	30–60	$ 20,441.46	0.02	$ 408.83	
6	>60	$ 33,193.55	0.3	$ 9,958.07	
7	Grand Total	$ 83,734.30			

STEP 9: Enter the sum formula in the next cell (D7 in this example solution):

=sum(D5:D6)

This sum is the desired adjusted year-end credit balance of allowance for uncollectible accounts. This total is $10,366.89 (also shown in Illustration 4.37).

STEP 10: Compare the 2025 unadjusted allowance account balance, a $3,000 credit balance, to the calculated desired ending balance to calculate the amount needed for the adjusting entry to ensure the allowance for uncollectible accounts has that desired credit balance:

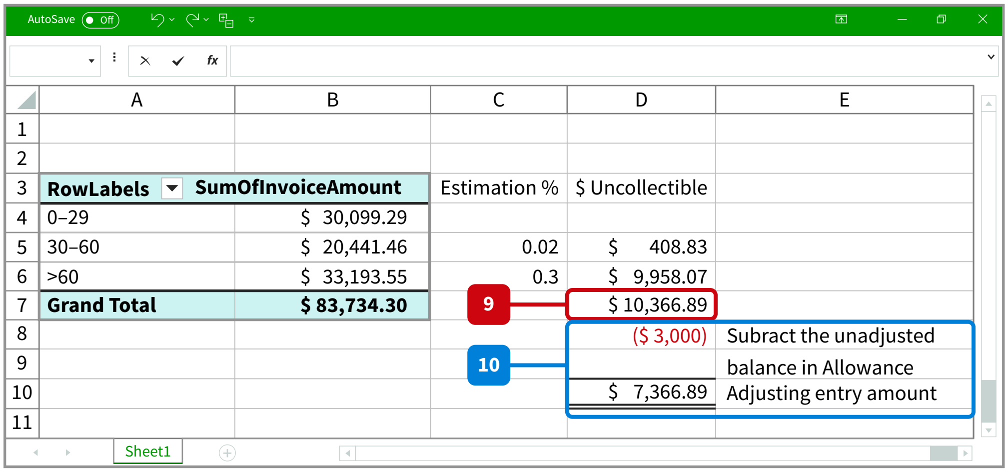
- Enter the unadjusted balance in cell D8 and the formula for the calculation in cell D10 (**Illustration 4.37**):

=D7−D8

- You are now ready to make your adjusting entry:

	Debit	Credit
Bad debt expense	7,366.89	
Allowance for uncollectible accounts		7,366.89

ILLUSTRATION 4.37 Completed Analysis of Bad Debts Expense Calculation



	A	B	C	D	E
1					
2					
3	RowLabels	SumOfInvoiceAmount	Estimation %	$ Uncollectible	
4	0–29	$ 30,099.29			
5	30–60	$ 20,441.46	0.02	$ 408.83	
6	>60	$ 33,193.55	0.3	$ 9,958.07	
7	Grand Total	$ 83,734.30	9	$ 10,366.89	
8				($ 3,000)	Subract the unadjusted
9			10		balance in Allowance
10				$ 7,366.89	Adjusting entry amount
11					

HOW TO 4.2 Create a Frequency Bar Chart in Power BI

How To

Illustration 4.16 can be recreated in Power BI by analyzing the outstanding invoices in accounts receivable by customer and creating a visualization that shows how many outstanding invoices are owed by unique customers at the end of 2024 and 2025 in three groupings: 1-3 invoices, 4-6 invoices, and 7-9 invoices.

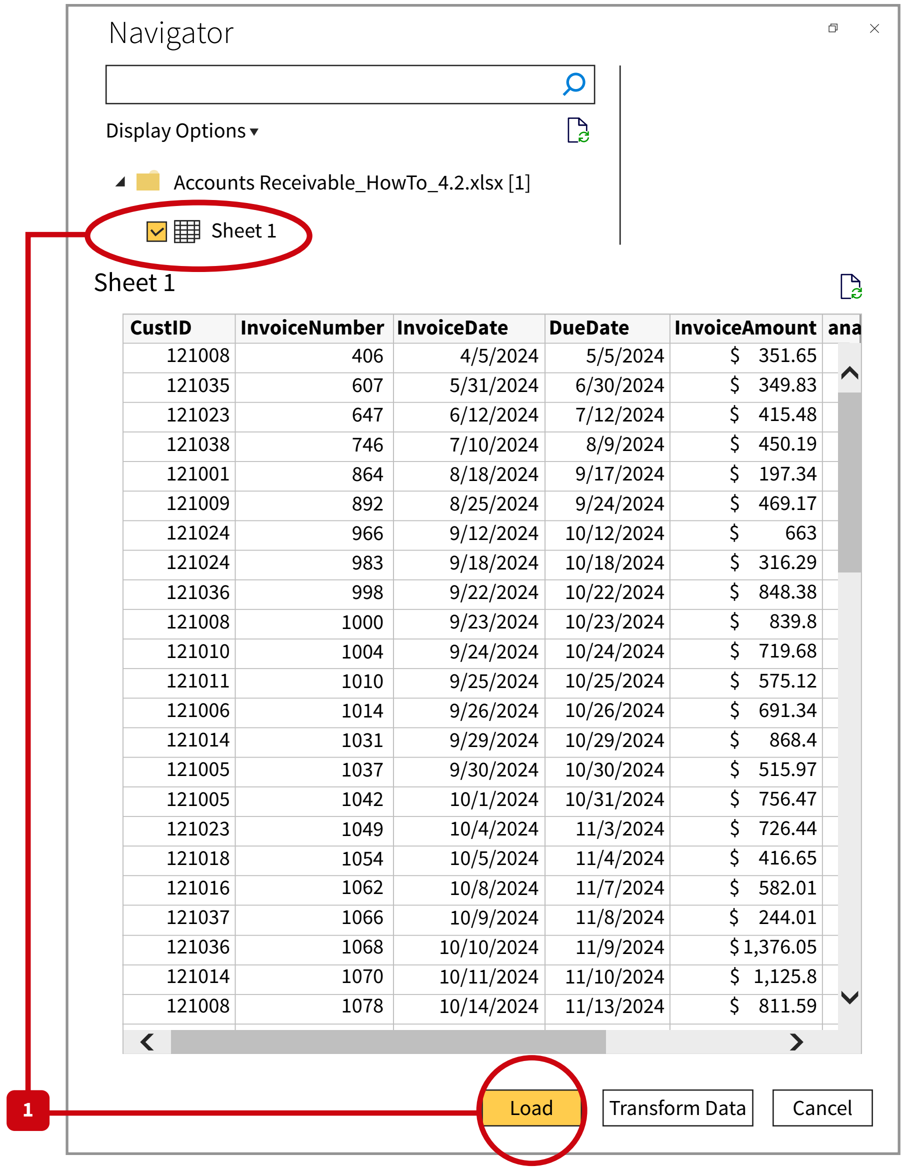
What You Will Need: Data The How To 4.2 data file.

STEP 1: Upload the data set into Power BI:

- Open Power BI and Select **Get Data**.
- Select **Import data** from Excel and choose **Excel Worksheet**.
- Navigate to the saved the data set file and open it.

Once the Navigator window opens, select Sheet 1 of the file listed on the left, then select **Load** (**Illustration 4.38**).

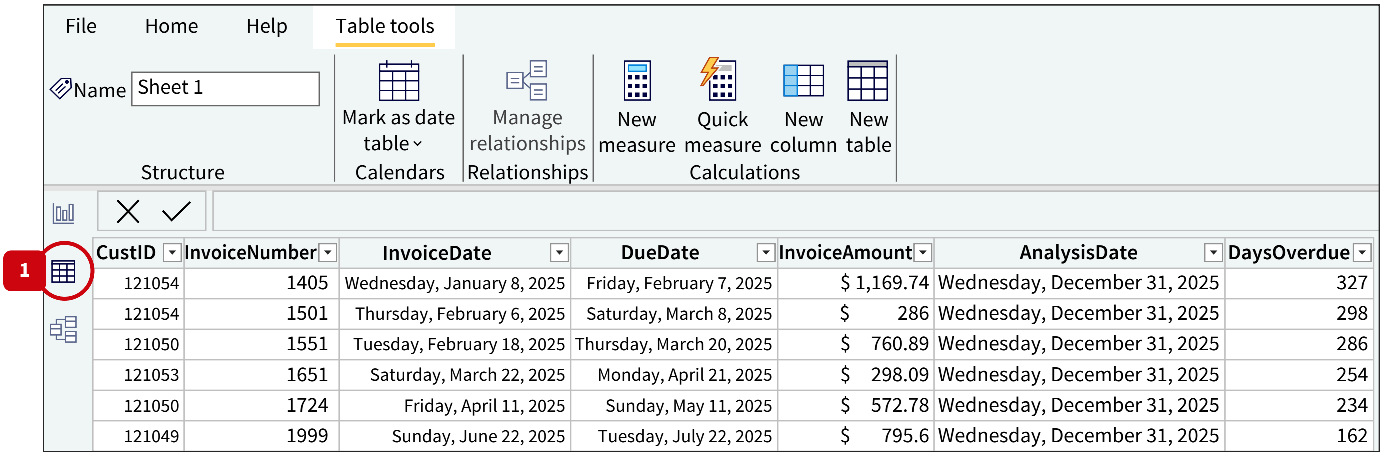
ILLUSTRATION 4.38 Load Data into Power BI



CustID	InvoiceNumber	InvoiceDate	DueDate	InvoiceAmount	ana
121008	406	4/5/2024	5/5/2024	$ 351.65	
121035	607	5/31/2024	6/30/2024	$ 349.83	
121023	647	6/12/2024	7/12/2024	$ 415.48	
121038	746	7/10/2024	8/9/2024	$ 450.19	
121001	864	8/18/2024	9/17/2024	$ 197.34	
121009	892	8/25/2024	9/24/2024	$ 469.17	
121024	966	9/12/2024	10/12/2024	$ 663	
121024	983	9/18/2024	10/18/2024	$ 316.29	
121036	998	9/22/2024	10/22/2024	$ 848.38	
121008	1000	9/23/2024	10/23/2024	$ 839.8	
121010	1004	9/24/2024	10/24/2024	$ 719.68	
121011	1010	9/25/2024	10/25/2024	$ 575.12	
121006	1014	9/26/2024	10/26/2024	$ 691.34	
121014	1031	9/29/2024	10/29/2024	$ 868.4	
121005	1037	9/30/2024	10/30/2024	$ 515.97	
121005	1042	10/1/2024	10/31/2024	$ 756.47	
121023	1049	10/4/2024	11/3/2024	$ 726.44	
121018	1054	10/5/2024	11/4/2024	$ 416.65	
121016	1062	10/8/2024	11/7/2024	$ 582.01	
121037	1066	10/9/2024	11/8/2024	$ 244.01	
121036	1068	10/10/2024	11/9/2024	$ 1,376.05	
121014	1070	10/11/2024	11/10/2024	$ 1,125.8	
121008	1078	10/14/2024	11/13/2024	$ 811.59	

- Verify this import by double-clicking on the data sheet icon on the left side of the screen. Look at the bottom left corner to verify that you have successfully loaded all 201 records of the data set (**Illustration 4.39**).

ILLUSTRATION 4.39 Verify Excel Import



CustID	InvoiceNumber	InvoiceDate	DueDate	InvoiceAmount	AnalysisDate	DaysOverdue
121054	1405	Wednesday, January 8, 2025	Friday, February 7, 2025	$ 1,169.74	Wednesday, December 31, 2025	327
121054	1501	Thursday, February 6, 2025	Saturday, March 8, 2025	$ 286	Wednesday, December 31, 2025	298
121050	1551	Tuesday, February 18, 2025	Thursday, March 20, 2025	$ 760.89	Wednesday, December 31, 2025	286
121053	1651	Saturday, March 22, 2025	Monday, April 21, 2025	$ 298.09	Wednesday, December 31, 2025	254
121050	1724	Friday, April 11, 2025	Sunday, May 11, 2025	$ 572.78	Wednesday, December 31, 2025	234
121049	1999	Sunday, June 22, 2025	Tuesday, July 22, 2025	$ 795.6	Wednesday, December 31, 2025	162

STEP 2: Select the **AnalysisDate** column.

- In the menu under **Column Tools**, pull down the **Format Field** options and choose the last option "2001 (yyyy)" (**Illustration 4.40**).
- Verify the analysis date field has changed to show 2024 and 2025.

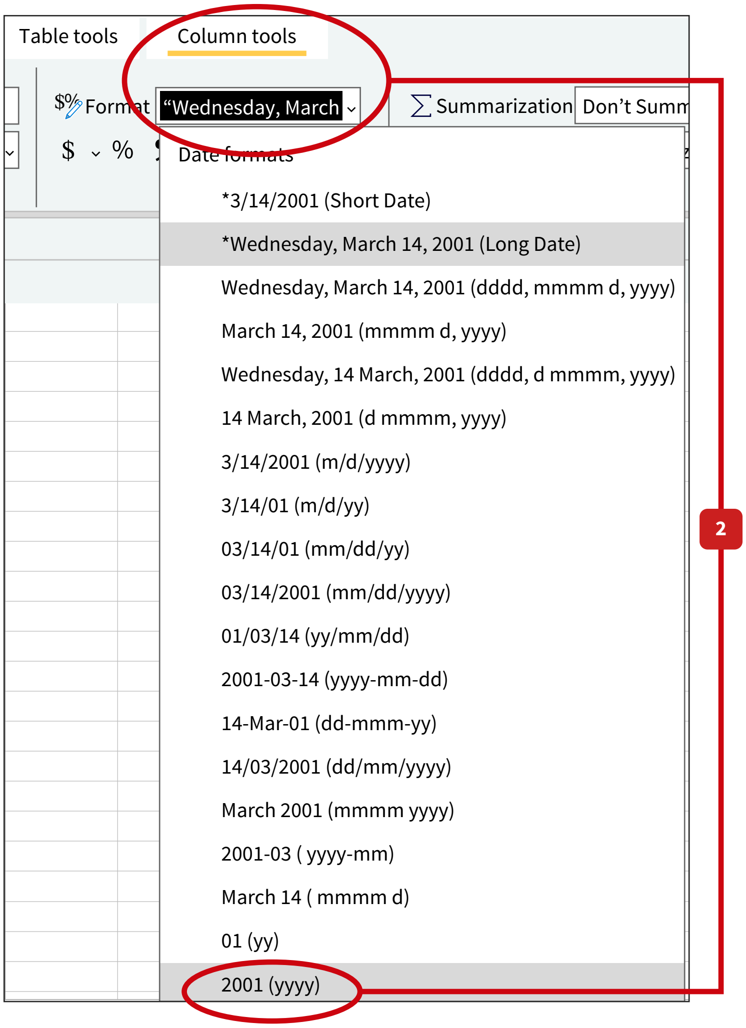


ILLUSTRATION 4.40 Change the AnalysisDate Column Format

STEP 3: Select the **CustID** column by clicking on the column header. Right click and select **Edit Query** (**Illustration 4.41**).

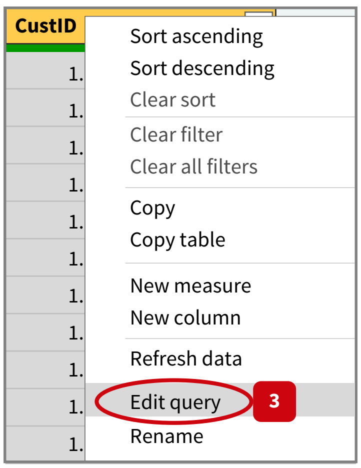
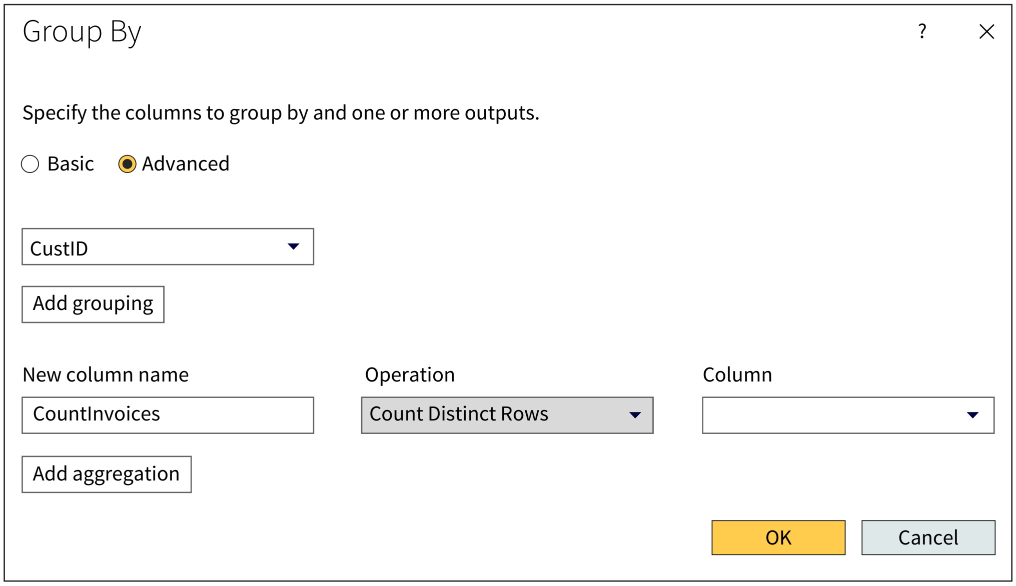


ILLUSTRATION 4.41 Edit Query Selection

- This will open the Power Query Editor window. In this window, the **CustID column** will also be selected.
- Select **Group By** from the top menu, and it will open.
- Select **Advanced**. Right below the CustID field you will see the Add grouping button. Select this to add a second grouping for the field AnalysisDate. Then, below the new grouping box, in the New column name box, enter "CountInvoices" (do not enter the quotes).
- In the pull down menu for **Operation**, select **Count Distinct Rows** and select **OK**. Close and Apply the Power Query Editor (**Illustration 4.42**).

ILLUSTRATION 4.42 Group By Customer ID and Count Distinct Invoices



STEP 4: Select the **Count Invoices** column. Select **Data Groups** from the top column, then select **New Data Group** (**Illustration 4.43**).

ILLUSTRATION 4.43 Select Data Groups

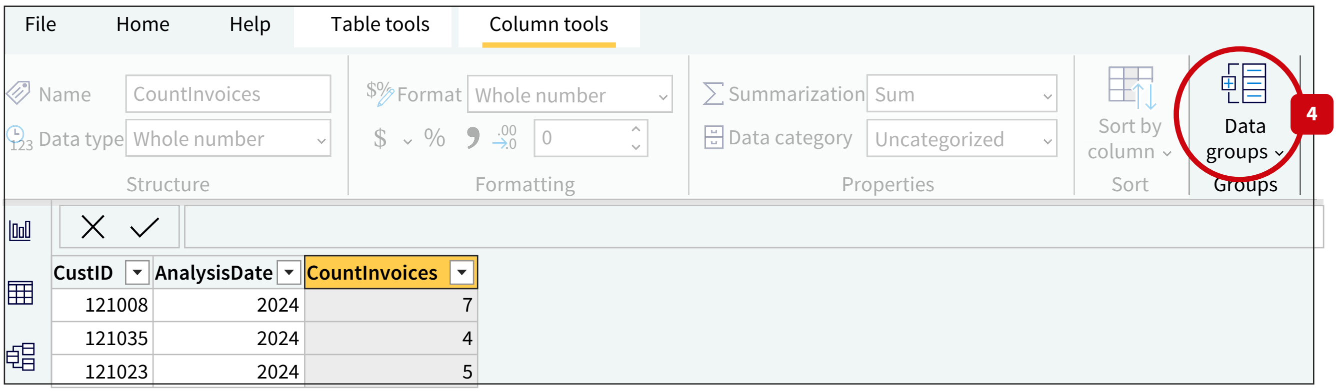
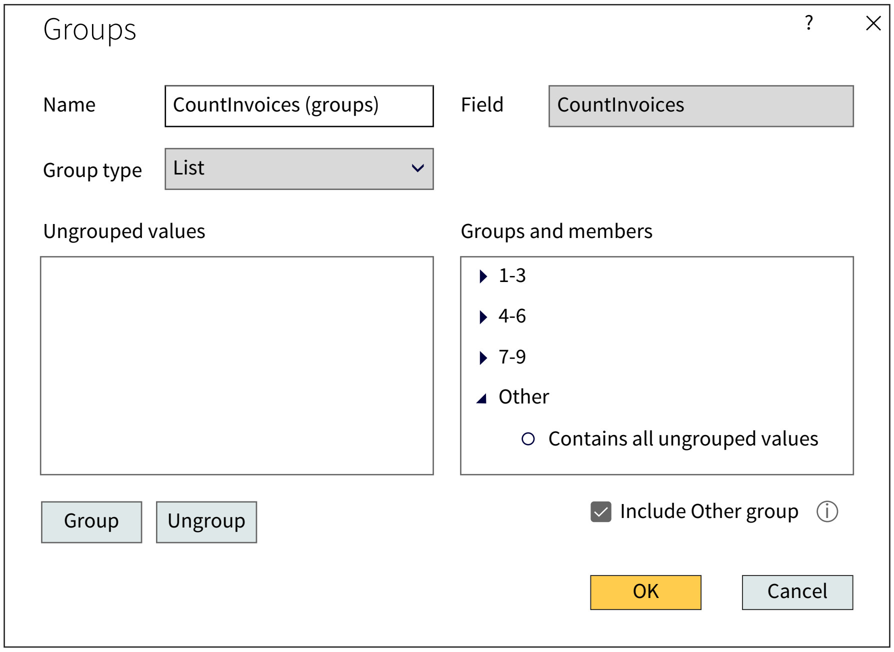


Illustration 4.43 shows the dialog box that opens once you select **Data Groups**:

- In the dialog box select **List** (not bins) for the Group Type.
- Under **Ungrouped values**, hold the shift key down to select values 1 2, and 3.
- Select **Group**. Double click the new group that appears on the right window and change the label to "1-3".
- Select the **Other** group, and move to the ungrouped left side. Select 4 and 5 together by again holding down the shift key. Select **Group** and move over to the right window. Double click on the label for this field, and type "4-6" (it happens that there are no customers with six outstanding invoices, but the range will be 4-6 regardless).
- Once again, select **Other** and move to the **Ungrouped values** window. Select 7, 8, and 9 by holding down the shift key and selecting **GROUP**.
- Go back to the right and double click on the label for this group. Change it to "7-9." Select **OK**. (**Illustration 4.44**).

ILLUSTRATION 4.44 Create Grouping of Number of Invoices

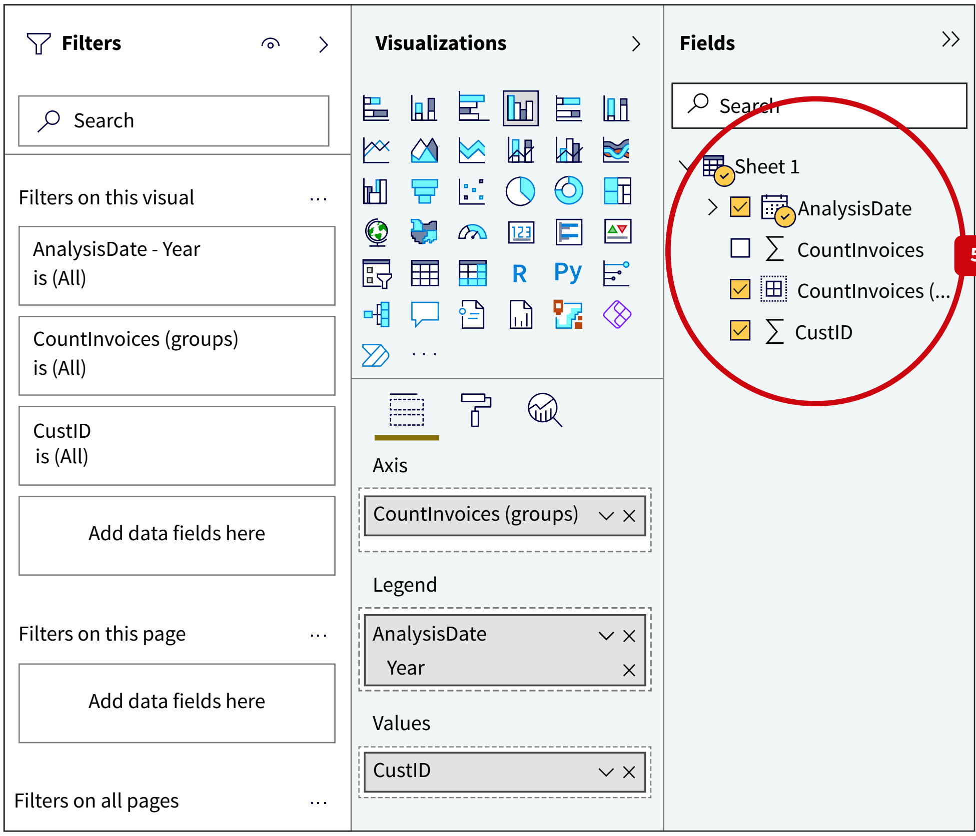


STEP 5: Now you are ready to create your visualization:

- Select the graphing icon on the left of the screen.
- Select all three variables in your **Fields** list.

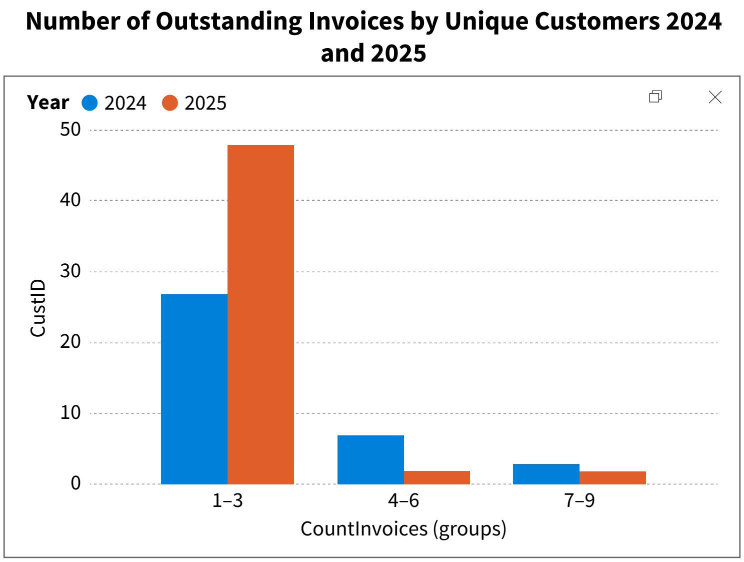
In the **Visualizations** pane, **Count Invoices** (groups) should be in the axis area. You may need to move the **AnalysisDate** field up to the **Legend** field by dragging it. The **Count Invoices** field should be in the **Values** area (**Illustration 4.45**).

ILLUSTRATION 4.45 Create the Visualization



Your visualization is complete (**Illustration 4.46**)!

ILLUSTRATION 4.46 WeMakeIt, Inc. Accounts Receivable



Data The Data tag appears when the data required to answer a question or complete an exercise are available on Wiley's online learning platform.

Multiple Choice Questions

1. **(LO 1)** Which of these is *not* an objective of data analysis strategies?
   - **a.** Collecting data.
   - **b.** Describing data.
   - **c.** Diagnosing data.
   - **d.** Predicting a future value of data.
   - **e.** Prescribing the impact of a particular strategy on the future value of the data.

2. **(LO 1)** Diagnosing objectives in data analysis strategies involve which of the following?
   - **a.** Looking for complete independence of variables.
   - **b.** Looking for causality in regression variables.
   - **c.** Looking for where to place the blame for bad operational results.
   - **d.** Looking for associations between data variables.
   - **e.** Predicting what future outcomes will be in the data.

3. **(LO 1)** The components of an effective data analysis strategy plan consider each of the following *except* for
   - **a.** analysis controls.
   - **b.** analysis risks.
   - **c.** analysis objectives.
   - **d.** analysis complexity.

4. **(LO 1)** Which of the following specific questions depicts a descriptive analysis objective for accounts receivable balance at year-end?
   - **a.** How many invoices from the current year were collected during the year?
   - **b.** What was the total of all sales revenues, cash and credit sales, for the year?
   - **c.** How many salespersons were employed in New York City this year?
   - **d.** What is the average days in inventory for each of your product lines?
   - **e.** How many of the open invoices at the end of the year are past due?

5. **(LO 1)** You are asked to perform an analysis predicting the accounts payable balance at the year-end. Which of the following data sources are best to analyze for this objective?
   - **a.** A list of authorized new vendors for the year.
   - **b.** Open vendor invoices that refer to purchases already received.
   - **c.** A list of checks paid to accounts payable vendors during the year.
   - **d.** A list of purchase orders issued during the month of December.
   - **e.** None of these data resources are useful to predict the accounts payable year-end balance.

6. **(LO 1)** Assume you are asked to estimate the amount of uncollectible receivables at year-end. Which of the following is *not* an analysis risk to consider when estimating accounts receivable bad debts at year-end?
   - **a.** The number of full-time sales employees is reported inaccurately.
   - **b.** The calculation of days outstanding is inaccurate.
   - **c.** The company changed collection policies for certain customers.
   - **d.** There are new customers who are not included in the analysis of aged receivables.
   - **e.** There are mathematical errors in the summation of receivables.

7. **(LO 1)** Which of the following is an appropriate response to the data risk that the sample selected for analysis is inappropriate?
   - **a.** Add missing variables to see explanatory power changes.
   - **b.** Test a sample distribution form for statistical test assumptions.
   - **c.** Verify the representativeness of your sample.
   - **d.** Explain the rule used for outlier removal.
   - **e.** Know your outlier data point.

8. **(LO 1)** Critically thinking through the risks to a data analysis strategy during planning will help avoid
   - **a.** data selection biases.
   - **b.** judgment errors from lack of experience.
   - **c.** interpretation errors from dirty data sets.
   - **d.** incorrect applications of statistical tests.
   - **e.** All of the these are possible costs from not considering the risks to data analyses choices.

9. **(LO 1)** A data analyses strategy is more likely to be successful if you critically think through
   - **a.** the analysis performed by your competitors.
   - **b.** the results you want to achieve.
   - **c.** the data that will best inform your questions.
   - **d.** your objectives for the analysis.
   - **e.** None of these help you perform better data analyses strategy planning.

10. **(LO 2)** Which of the following refers to the content of a data set's row?
    - **a.** Data field.
    - **b.** Attribute.
    - **c.** Single occurrence of the data record.
    - **d.** All these define a row of data in a dataset.

11. **(LO 2)** Which of the following could refer to the name of a column in a sales invoices data set?
    - **a.** Invoice 3570.
    - **b.** Customers in Orlando, Florida.
    - **c.** CustomerState.
    - **d.** BankStatementDate.

12. **(LO 2)** Which of the following is an example of knowledge you might need to acquire for data analysis strategy planning?

- **a.** The analysis is appropriate for your data measurement scales.
- **b.** An understanding of your biases.
- **c.** Identifying the key stakeholders.
- **d.** Analysis resources that are available for your compensation.
- **e.** Compensation correlation with results.

13. **(LO 2)** Which of the following is *not* a data measure scale that can be used in designing your data analysis strategy?

- **a.** Optimal data
- **b.** Ordinal data
- **c.** Interval data
- **d.** Ratio data
- **e.** All of these are data measure scales.

14. **(LO 2)** Which of the following data measure scales allows for meaningful multiplication of data points?

- **a.** Interval data
- **b.** Categorical data
- **c.** Ordinal data
- **d.** Ratio data
- **e.** Group data

15. **(LO 3)** Descriptive analyses that include median central tendency measures are appropriate for which data measurement scale?

- **a.** Categorical
- **b.** Ordinal
- **c.** Nominal
- **d.** Interval
- **e.** All these measurement scales support the median central tendency measure.

16. **(LO 3)** Diagnostic analyses that include data dispersion measures such as variance and standard deviation are appropriate for which data measurement scale?

- **a.** Categorical
- **b.** Ordinal
- **c.** Ratio
- **d.** Nominal
- **e.** All measurement scales are appropriate for variance analysis.

17. **(LO 3)** Bar chart visualizations are appropriate for data with which measurement scale?

- **a.** Categorical
- **b.** Ordinal
- **c.** Ratio
- **d.** Nominal
- **e.** All measurement scales are appropriate for bar charts.

18. **(LO 3)** Which of the following objective questions reflects a diagnostic purpose for accounts receivable year-end balance?

- **a.** How many outstanding invoices are there in the year-end accounts receivable balance?
- **b.** What is the value of the year-end accounts receivable balance?
- **c.** Are new customers causing the increase in outstanding invoices in the year-end accounts receivable balance compared to last year?
- **d.** What is the median value for outstanding invoices in the year-end accounts receivable balance?
- **e.** What is the invoice that has been outstanding the longest period of time in the year-end accounts receivable balance?

19. **(LO 3)** Predictive and prescriptive analyses that include creating trendline visualizations are appropriate for which data measurement scale?

- **a.** Categorical
- **b.** Ordinal
- **c.** Nominal
- **d.** Interval
- **e.** All of these measurement scales support using trendline visualizations.

20. **(LO 2, LO 3)** Which of these analysis strategies involve categorical and ratio data measurement scales?

- **a.** Reconciliations of subsidiary ledger balances to the general ledger balance.
- **b.** Depreciation schedules for all real estate assets.
- **c.** Which types of capital sourcing have the highest cost of capital.
- **d.** Changes to the tax rates applicable to their organization.

21. **(LO 4)** Managerial accountants would most likely plan data analyses when which of the following occur?

- **a.** Changes to the tax rates.
- **b.** Elimination of the financial statement footnote disclosure requirements.
- **c.** Changes to Securities and Exchange Commission website.
- **d.** Changes to their organization's performance.

22. **(LO 4)** AIS accountants would most likely design a data analysis strategy for which of the following opportunities?

- **a.** New technologies that can provide cost savings for operational processes.
- **b.** New customers in the marketplace.
- **c.** Turnover in the regulatory body leadership.
- **d.** Changes to the footnote disclosures required by generally accepted accounting rules.
- **e.** Changes in tax rules.

23. **(LO 4)** Auditors plan data analyses strategies for which of the following situations?

- **a.** Evaluating the effectiveness of internal controls in their client's accounting system.
- **b.** Evaluating the representativeness of an account balance on their client's balance sheet or income statement.
- **c.** Considering the risk involved in providing an audit opinion for a client.
- **d.** Investigating whether a client's assumptions for recording their lease liabilities is consistent with other companies in their industry.
- **e.** All of these are examples of situations where auditors plan their data analysis strategies.

24. **(LO 4)** Financial accountants would most likely plan data analyses strategies for which of the following?

- **a.** Changes to the management team.
- **b.** New competitors in the marketplace.

c. Changes to their assumptions and estimates for end of period adjusting entries.

d. Changes to the tax rates applicable to their organization.

e. None of these would motivate data analyses by financial accountants.

25. **(LO 4)** Tax accountants would most likely plan a data analysis strategy that can

a. describe cash outflows for business expenses during the last year.

b. evaluate changes to the financial statement footnote disclosure rules.

c. compare advertising expenses to sales revenue increases.

d. compare organizational strategy options to predicted operational performance.

e. None of these would motivate data analyses by tax accountants.

Review Questions

1. **(LO 1)** Describe the five components in a data analysis project plan.

2. **(LO 1)** Discuss why it is important for accounting professionals to focus on the objective and questions to generate data and analysis alternatives.

3. **(LO 1)** Describe data and analysis risks that should be considered when planning a data analysis project.

4. **(LO 2)** Define and describe the four measurement scales used as accounting data variables.

5. **(LO 2)** Explain how selecting certain data fields could be inappropriate for your project plan.

6. **(LO 2)** Provide examples of two common data risks and a control for each that would reduce the risk.

7. **(LO 3)** Discuss why it is important to know the measurement scales of variables used in the analysis when determining whether to design a descriptive, diagnostic, predictive or prescriptive data analysis strategy.

8. **(LO 3)** Discuss why it is important to match the type of analysis performed with the measurement scale of the data.

9. **(LO 3)** Explain a trend analyses strategy, and provide an example of how an accountant may analyze sales using a trend analysis.

10. **(LO 4)** Assume you are a systems accountant working in the accounting information systems group at your company. You have been asked to perform data analysis to analyze the number of spam emails that pass through to employees before and after the company's investment in a new firewall. Discuss data risks and identify related controls to minimize those risks in your analysis.

11. **(LO 4)** Assume you are an auditor working to test internal controls regarding p-card usage at a company. You have the following data fields:

EmployeeNumber

EmployeeName

P-cardTransactionDate

P-cardTransaction vendor

P-cardTransactionAmount

Describe a descriptive analysis strategy you can perform using the data strategy of the p-card transaction amount and any one of the other data fields.

12. **(LO 4)** Assume you are a tax accountant evaluating an individual's financial records for their federal tax compliance. What data risks should be considered?

Brief Exercises

BE 4.1 **(LO 1)** Put the data analysis project plan components in their proper sequential order, starting with number 1 for the first step.

___ **a.** Embed controls.

___ **b.** Identify project objective and specific questions.

___ **c.** Design data strategy.

___ **d.** Consider risks.

___ **e.** Design analysis strategy.

BE 4.2 **(LO 1)** Put the steps of creating a data strategy in proper order, starting with number 1 for the first step.

___ **a.** Choose and value the factors upon which to rank the data alternatives.

___ **b.** Evaluate data alternative rankings and select the best data strategy option.

___ **c.** Based on the objective, develop a few data alternatives.

___ **d.** Extract, clean, and transform the data fields to prepare the data for analysis.

BE 4.3 **(LO 1)** **Financial Accounting** For each of the following, list which critical thinking element is most clearly involved (S = Stakeholders; P = Purpose; A = Alternatives; R = Risks; K = Knowledge):

___ **1.** Knowing what valuation procedure to use for U.S. GAAP.

___ **2.** Considering the impact of the analysis results.

___ **3.** The basis for the appropriateness of your data selections.

___ **4.** Not just going with the first idea that you think of for your data and your analysis.

___ **5.** Performing an analysis that is not appropriate for the measurement scale of your data.

BE 4.4 **(LO 2)** Match the data measurement scale that could be utilized to each example.

Measurement scales can be used once, more than once, or not at all.

a. Categorical
b. Ordinal
c. Interval
d. Ratio

Example	Measurement Scale
1. Customer survey ratings.	
2. Account numbers in the chart of accounts.	
3. States or regions of the country.	
4. Book value of depreciable assets.	
5. Transaction dates, such as the dates orders are shipped.	
6. Credits scores of loan applicants.	
7. The VendorID field.	
8. Sales revenues.	
9. The set weight of product boxes shipped.	
10. Supermarket inventory product type, such as produce, pet food, meat, and frozen foods.	

BE 4.5 **(LO 2)** **Auditing** Assume you are an audit senior responsible for supervising this year's intern on a data analysis project related to the client's purchases. You provided the intern with a data set and requested that the intern review the data set and be prepared to discuss. An excerpt of the data follows.

PONo	VendID	VendorName	Vendor Quality	Vendor Payterms	PODate	PO ItemNo	ItemCost	ItemQty
2001	783	Pep N Supply	3	1/10 net 30	12/1/2023	23568	41.99	12
2002	783	Pep N Supply	3	1/10 net 30	12/2/2023	23567	31.99	12
2003	784	Playtime Toys	4	1/15 net 30		32425	15.99	11
2004	784	Playtime Toys	4	1/15 net 30	12/4/2023	32426	15.99	10
2005	784	Playtime Toys	4	1/15 net 30	12/5/2023	32427	15.99	25
2006	258	Snappy Supplies	5	net 45	12/6/2023	11246	3.25	26
2007	153		2	2/10 net 60	12/7/2023	11258	333.25	35
	153	Production	2	2/10 net 60	12/8/2023	11259	6.75	45
2009	783	Pep N Supply	3	1/10 net 30	12/9/2023	23566	22.99	100

1. The intern stated, "Reviewing the data set, I classified VendorID as a measured raw data field source." Explain why VendorID should be considered non-measured raw data.
2. The intern stated, "The VendorQuality data field must be considered ratio data because we can analyze this as a variable in our analysis." Explain why the VendorQuality data field should not be considered ratio data.
3. The intern said, "The ItemCost and ItemQty data fields can be combined to create a calculated data field." Explain why this is correct.
4. Based on your review of the excerpt of the data set, what data risks would you expect the intern to identify?
5. Assume you asked the intern to verify the accuracy of the PO number 2006. What would the intern review to determine if the data in the database is accurate?

BE 4.6 **(LO 2)** For each of the following, identify if the data field is a (MRD) measured raw data field, a (NRD) non-measured raw data field, or a (CAL) calculated field.

___ **1.** ProductIdentificationCode

___ **2.** UnitCost for each inventory item

___ **3.** TotalCost

___ **4.** GrossProfit

___ **5.** ProductCategory

___ **6.** NumberOnHand in the warehouse of each product

BE 4.7 **(LO 3)** Identify the objective of each of the following analysis strategies as describe, diagnose, predict, or prescribe.

Strategy	Objective
1. Aging of accounts receivable's year-end open invoices on days outstanding to understand what is happening in accounts receivable.	
2. Estimating cost savings and productivity increases from new AIS investments.	
3. Creating dollar value stratified sampling plans for investment asset accounts.	
4. Counting and ranking web server security incidents by week.	
5. Estimating product revenue increases by product from marketing campaigns.	
6. Estimating future cash flows from a continuing current operations strategy.	
7. Calculating before-after cost reductions from system investments.	
8. Testing different tax planning strategies by evaluating future cash savings.	
9. Estimating inventory counts at year-end by sampling a few of the stores and making population inferences.	
10. Frequency counts for unfulfilled or open-partial sales orders so that they can be resolved.	

BE 4.8 **(LO 3)** For each data structure and analysis listed for descriptive and diagnostic objectives, indicate whether the analysis can be performed with a "Yes" or "No."

___ **1.** The calculation of the mode on categorical data.

___ **2.** The variance dispersion of ordinal data.

___ **3.** The calculation of median on interval data.

___ **4.** Cumulative frequency distributions on categorical data.

___ **5.** A visualization of categorical data trends.

___ **6.** Cross tabulation correlations of ratio data.

BE 4.9 **(LO 3)** For each data structure and analysis listed for predictive and prescriptive objectives, indicate whether the analysis can be performed with a "Yes" or "No."

___ **1.** Trendlines on categorical data.

___ **2.** Regression analysis of ordinal data.

___ **3.** Optimization modeling of interval data.

___ **4.** Regression analysis of categorical data.

___ **5.** What-if analysis of ratio data.

___ **6.** Trendlines on ratio data.

BE 4.10 **(LO 4)** **Auditing** For each of the following data strategies used by auditors, provide at least one risk that should be controlled to rely on the results of the analysis.

1. Sampling the clients' transactions by the size of the transactions impacting an account.
2. Comparing the estimations and assumptions used by management this year to last year.
3. Requesting confirmations of all year-end bank account, accounts receivable, and accounts payable balances.

BE 4.11 **(LO 4)** **Accounting Information Systems** For each of the following analyses strategies for AIS projects, identify whether the project's objective is descriptive or diagnostic (D) or predictive or prescriptive (P):

Strategy	Objective
1. Calculating the average amount of time each department in the revenue cycle takes to perform its function per order.	
2. Calculating the cost savings expected from moving to RFID (radio free identification) technologies for inventory identification, pricing, and costing.	
3. Estimating how much server space would be needed for the AIS if sales revenues were to increase by 10% each year for the next five years.	
4. Detecting at which times of day and by which channel the unauthorized access is occurring to the customer relationship management module.	

BE 4.12 **(LO 4)** **Financial Accounting** Assume you are a financial accountant tasked with selecting the best financing alternative for your organization. Suggest a data strategy and an analysis strategy for this objective.

Exercises

EX 4.1 **(LO 1)** **Match Project Plan Components** Match the data analysis project plan component to each of the following statements:

a. Project objectives and questions
b. Data strategy
c. Analysis strategy
d. Critical risks
e. Embedded controls

___ **1.** Verifying data values for reasonableness.
___ **2.** Data formatting that must be transformed before analysis.
___ **3.** Predicting next year's sales volume.
___ **4.** Sales volume data.
___ **5.** Using regression analysis to predict next year's sales volume.
___ **6.** Diagnosing what is causing the cost overruns in the production process.
___ **7.** Use of different scales on the y-axis and x-axis across visualizations.

EX 4.2 **(LO 1)** **Managerial Accounting** **Financial Accounting** **Design a Data Strategy and an Analysis Strategy** Assume your motivation for performing data analysis is to maximize sales revenues. Your specific objective questions are to determine whether sales volume is increasing, stable, or dropping for each of your company's product lines this year compared to last year. Describe one possible data strategy and analysis strategy for this objective.

EX 4.3 **(LO 1)** **Managerial Accounting** **Accounting Information Systems** **Match Data Strategy Choices** Match each objective to the following data selections.

a. Labor cost data
b. Production run data
c. Past marketing strategy data
d. IT service satisfaction survey data
e. Online market sales data
f. Sales contract data

___ **1.** What is causing our large production materials usage variance?
___ **2.** Which marketing strategy is most likely to increase our market share?
___ **3.** Are we in compliance with U.S. GAAP procedures for revenue recognition?
___ **4.** Which of our product lines could be sold online as well as in the store?
___ **5.** Is the turnover in our IT staff impacting their service to the other functional areas?
___ **6.** Are our payroll costs increasing?

EX 4.4 (LO 2) Managerial Accounting Design Data Strategies The following is a list of the available fields in a data set from a pet supply store.

Data Field	Description
1. PONo	Uniquely assigned purchase order number
2. VendorID	Uniquely assigned vendor number
3. VendorName	Vendor name
4. VendorQuality	Quality rating ranging from 1 - 6 where 1 = poor and 6 = exceptional
5. VendorAddress	U.S postal mailing address and street name
6. VendorCity	Mailing address city
7. VendorState	Mailing address state
8. VendorZip	Mailing address zip
9. VendorPayterms	Payment terms negotiated with the vendor
10. PODate	Purchase order date
11. POItemID	Uniquely identifiable item number
12. POItemDescription	Description of item
13. ItemCost	Cost of the inventory item per negotiated terms with vendor
14. ItemQty	Total quantity of items purchased

For each of the following objectives (a, b, c), identify the minimum data fields (1-14) you would select for your data strategy:

a. Your objective is to reduce the carbon footprint of your purchases, so you want to identify which vendor transported your purchases across the most miles last year.

b. Your objective is to determine which inventory items have the greatest cost increases over the past five years.

c. Your objective is to identify the five vendors from whom you made the highest value of purchases last year.

EX 4.5 (LO 2) Prepare a Project Plan You graduate in one month from your undergraduate program. Your objective is to successfully apply to both professional jobs and graduate school. In your application materials, you want to show how your grades improved as you moved from your general education courses to the courses in your major. Describe your data strategy, your analysis strategy, and three possible risks you need to control by providing control suggestions.

EX 4.6 (LO 2) Managerial Accounting Data Strategy Risks and Controls Assume your data analysis objective is to reduce the carbon footprint of your purchases from vendors. Your analysis strategy is to identify which vendor transported your purchases across the most miles last year. You have selected the following data fields for your data strategy.

Data Field	Description
PONo	Uniquely assigned purchase order number
VendorID	Uniquely assigned vendor number
VendorName	Vendor name
VendorAddress	U.S postal mailing address and street name
VendorCity	Mailing address city
VendorState	Mailing address state
VendorZip	Mailing address zip
PODate	Purchase order date

Identify three data risks and data controls you could use for those risks.

EX 4.7 (LO 3) Auditing Managerial Accounting **Prepare a Project Plan** You are an internal auditor working with your company's outdoor gear sales department to devise a new strategy for sales commissions that incentivizes higher profits than your current flat commission strategy. Describe a data strategy, an analysis strategy and three possible risks you need to control by providing control suggestions.

EX 4.8 (LO 3) Managerial Accounting **Prepare a Project Plan** You are a managerial accountant for Sihrya's Beauty Supply Store. You have been asked to predict next year's contribution margin. A sample of the available data for your analysis is included here.

Field Label	Field Name in Database
Receipt Number	SaleReceiptNo
Sales Date	Saledate
Inventory Code	InvCode
Number Sold	NoSold
Inventory Description	InvDesc
Inventory Price	InvPrice
Inventory Cost	InvCost

Describe a data strategy, an analysis strategy, and three possible risks you need to control by providing control suggestions.

EX 4.9 (LO 3) Financial Accounting Managerial Accounting **Prepare a Project Plan** You work as a financial analyst for a large coffee shop chain, such as Starbucks. Your objective is to diagnose which stores have increasing sales. Your data strategy is to use the store sales monthly data from the past two years. Design an analysis strategy, including which risks you should consider and controls that would help mitigate those risks.

EX 4.10 (LO 1- 4) Auditing Accounting Information Systems **Prepare a Project Plan** You are working in the accounting information systems group at your company. You have been asked to evaluate the internal controls associated with authentication of users into the information system. Your manager has asked you to perform a descriptive analysis of failed login attempts. Your manager has provided you with the analysis objective and question. Using the chart, document your data and analysis choices and risks and controls for each.

Objective and Questions	Data and Analysis Strategies	Risks	Controls
Objective: Evaluate internal controls associated with authentication of users	**1.** Data:	**3.** Data:	**5.** Data:
Questions: What are the mean, median, standard deviation, and distribution of failed login attempts during the current year?	**2.** Analysis:	**4.** Analysis:	**6.** Analysis:

EX 4.11 (LO 1-4) Financial Accounting Managerial Accounting Prepare a Project Plan You are a financial analyst at Sihrya's Beauty Salon. The company owner has asked you to perform data analyses to understand the products that contribute to the salon's retail store's profitability. The owner provided you a data dictionary, presented here, describing data that you may consider using in your analysis.

Field Label	Field Name in Database	Field Description
Receipt Number	ReceiptNo	POS assigned receipt number uniquely identifying each sale.
Sales Date	SaleDate	Date of sale per the POS.
Inventory Code	InvCode	Unique inventory identification number for each product in the salon's retail store.
Number Sold	NoSold	Quantity of items sold.
Inventory Description	InvDesc	Description of the inventory item.
Inventory Price	InvPrice	Gross sales price of the inventory item.
Inventory Cost	InvCost	Weighted average cost of the inventory item.

1. State the owner's objective for your data analysis project.
2. Assume your analysis question is to identify the products that have the highest gross profit. Identify the data fields you should include in your analysis. Identify the data analysis choices to answer this analysis question.
3. Identify risks and the related control choices related to the analysis question.

EX 4.12 (LO 1-4) Data Managerial Accounting Select a Data Strategy and Perform an Analysis You are a managerial accountant working at a multi-location pet care retail company. Your supervisor has asked you to compare the dollar amount of purchases made from each vendor in December 2024 compared to December 2025. After discussing with your supervisor, you have identified the following:

Objective: Compare total purchases by vendor in December 2024 and December 2025.

Question: In 2024 and 2025, from which vendors were the most purchases (in dollars) made?

Data: You have access to the following data.

Data Field	Description
PONo	Uniquely assigned purchase order number.
VendorID	Uniquely assigned vendor number.
VendorName	Vendor name.
VendorQuality	Quality rating ranging from 1 - 6 where 1 = poor and 6 = exceptional.
VendorAddress	U.S postal mailing address and street name.
VendorCity	Mailing address city.
VendorState	Mailing address state.
VendorZip	Mailing address zip.
VendorPayterms	Payment terms negotiated with the vendor.
PODate	Purchase order date.
POItemID	Uniquely identifiable item number.
POItemDescription	Description of item.
ItemCost	Cost of the item per negotiated terms with vendor.
ItemQty	Total quantity of items purchased.

1. Which data items would you use in your data strategy?
2. Design an analysis strategy to identify the vendor or vendors in which the company had a higher dollar amount of purchases in 2025 compared to 2024.
3. Perform the analysis per your analysis strategy. Identify the vendors in which the company had a higher dollar amount of purchases in 2025 compared to 2024.

Problems

PR 4.1 (LO 1-4) Data Auditing Managerial Accounting Complete Project Plan You work in the internal audit group in your organization, and your supervisor has asked you to analyze p-card data. The objective of the analysis is to understand p-card spending in the current year. The questions associated with the objective include:

- With which three vendors does the company spend the most money using p-cards?
- Which employee spends the highest dollar amount of money using the p-card?

Review the data and complete the chart to document your data and analysis strategy choices.

Objective and Questions	Data and Analysis Strategies	Risks	Controls
Objective: Understand p-card spending in the current year. Questions: • What are the three vendors where the Company spends the most money using p-cards? • Which employee spends the highest dollar amount of money using the p-card?	Data: Use the data provided in the Excel file. Analysis: Use Excel to create a PivotTable that allows grouping of the p-card spending data by vendor and sort by the highest vendor. Use Excel to create a PivotTable to group p-card spending amount by employee and sort by the highest dollar amount by employee.	**1.** Data: **2.** Analysis:	**3.** Data: **4.** Analysis:

5. Perform the analyses suggested in the chart. Summarize your results.

PR 4.2 (LO 1, 2, 3) Data Financial Accounting Managerial Accounting Complete Steps 2 and 3 of a Project Plan and Perform Analysis. You are a financial analyst at Sihrya's Beauty Salon. The owner has asked you to perform data analyses to understand the products that contribute to the salon's retail store's profitability. The owner gave you a data dictionary, which is presented here, describing data that you may consider using in your analysis.

Field Label	Field Name in Database	Field Description
Receipt Number	ReceiptNo	POS assigned receipt number uniquely identifying each sale.
Sales Date	Saledate	Date of sale per the POS.
Inventory Code	InvCode	Unique inventory identification number for each product in the salon's retail store.
Number Sold	NoSold	Quantity of items sold.
Inventory Description	InvDesc	Description of the inventory item.
Inventory Price	InvPrice	Gross sales price of the inventory item.
Inventory Cost	InvCost	Weighted average cost of the inventory item.

Assume your analysis objective is to identify the products that contribute the most to the salon's retail store profitability. Your specific questions are: Which products have the highest units sold? Which products have a positive gross profit margin? Use the information provided to answer the following questions:

1. Identify the data fields to include in your data strategy.
2. Identify a data strategy that could be used to answer the objective questions.
3. Identify an analysis strategy that could be used to answer the objective questions.
4. Review the data provided to perform the analysis. After performing the analysis, identify the top two products that contribute to the salon's profitability.

PR 4.3 (LO 1-4) Data Managerial Accounting Complete the Project Plan and Perform the Analysis As a financial analyst at a hospitality industry company, you have been asked to design a data analysis strategy to understand the factors that influence a guest's quality rating. Your team administered a survey of guests staying at your properties during the month of June. The database team has linked the survey responses for quality rating of the stay to the guest's location, check-in and check-out date, and spending amounts. They indicated that they only included data related to surveys completed September 1 through September 10.

1. Which data measurement scales are included in the data fields labeled Location, QualityRating, and SpendingAmount?
2. Review the data set, identify data risks present in the data set, and suggest controls for these risks.
3. Assume a member of your team suggested that you use the data to predict the quality rating of future stays. To do this predictive analysis, you would have to use the quality rating data in a predictive analysis strategy. How would you respond to your team member about the appropriateness of using this type of analysis?
4. Use this data set to design and perform a different, original data strategy and an analysis strategy to satisfy the objective question. Document your data strategy, analysis strategy, risks, and controls and present your analysis results.

PR 4.4 (LO 1-4) Data Tax Accounting Complete the Project Plan and Perform the Analysis Beautiful Bites is a chain of bakeries located in Colorado. Its owners are committed to social sustainability values and want to focus their philanthropy in the municipalities where they are collecting the most sales tax. The objective of the data analysis is to determine where most of their customers live. The specific questions is: In which communities do most of our customers reside?

1. Prepare the data analysis project plan.
2. Perform the analysis and summarize the results.

Professional Application Case: Automated Transportation, Inc. (ATI)

Automated Transportation, Inc. is a medium-sized manufacturer of remote-controlled cars, boats, and drones. The company was established five years ago when two brothers decided to build and sell remote control cars to enthusiasts. As the company grew, they expanded their product offerings to include remote control cars, boats, and drones. The brothers serve as the president and CEO of the company, and they now have over 70 employees.

They have two key customer groups, hobbyists and businesses interested in incorporating drones into their business process and supply chain. These two customer bases provide a variety of opportunities for growth. Management values internal control, but since they are busy running the company they have employed accounting personnel to help design the company's processes, policies, and internal controls:

- The company is not required to have a report on internal controls and external auditors are not required to attest to the company's internal controls.
- However, the owners want to be sure the company employees are designing and following policies that will help them maintain efficient and effective operations.
- The owners are also very data-driven and make many of their business decisions only after considering the data collected and analyzed.

The following is an excerpt from the data dictionary designed by the company's information systems personnel and accounting information systems accountants.

Data Field Label	Field Name in Database	Field Description
Invoice number	InvoiceNO	The invoice number, which is hand keyed into the AIS by the AP clerk from a manual invoice mailed to the company by the vendor.
Invoice amount	InvoiceAmt	The amount of the invoice, which is hand-keyed into the AIS by the AP clerk from a manual invoice mailed to the company by the vendor.
Shipment date	ShipDate	The date the product was shipped from the shipping location.

(Continued)

Data Field Label	Field Name in Database	Field Description
Invoice date	InvoiceDate	The date of the invoice, which hand-keyed into the AIS by the AP clerk from a manual invoice mailed to the company by the vendor.
Vendor identification number	VendorID	Unique vendor identification number.
Vendor name	VendorName	Name of the vendor.
Product purchased	ProductID	Product code for the product purchased from the vendor. This product code is consistent with the catalog from the vendor.
Unit cost	UnitCost	The cost per unit of the product purchased from the vendor.
Shipping cost	ShipCost	Total shipping costs.
Flat duty	FlatDuty	The flat duty rate applies to article that are dutiable.
Tariff	TariffAmt	The total dollar amount of a tariff applied to the goods shipped.
Shipping location	ShipLocation	The country in which the goods are shipped from.
Receiving Quality rating	QualityRate	This is a quality rating keyed into the AIS by the receiving team when they receive the goods. The scale is 1 = poor to 5 = excellent quality. The receiving team rates the shipment on packaging, quality of materials, and overall delivery.
Payment terms	PaymentTerms	The agreed-upon payment terms with the vendor. These terms are negotiated by the purchasing manager and keyed into the vendor master file by the purchasing supervisor.
Shipping terms	ShipTerms	Shipping terms–typically FOB destination or FOB shipping.
Payment address	PayAddress	The vendors address where payment is to be mailed.
Purchase order number	PONumber	The unique identifying number assigned to each purchase order issued by the company.
Purchase order date	PODate	The date the purchase order is issued by the company.
Receiving report number	ReceivingNumber	The unique identifying number assigned to each receiving report created by the company's receiving group.
Receiving report date	ReceivingDate	The date the product is received by the company's receiving department.
Quantity received	QtyReceived	The total quantity of items received.
Quantity purchased	QtyPurchased	The total quantity of items on the Purchase Order.
Invoiced quantity	QtyInvoice	The total quantity of items on the invoice, this amount is hand keyed into the AIS by the AP clerk from the manual invoice mailed to the company by the vendor.
Treasurer Approval	Approved	The initials of the Treasurer indicating their approval if the invoice was greater than $10,000.

PAC 4.1 Accounting Information Systems: Evaluate AIS Internal Controls

Accounting Information Systems As an information systems accountant, the owners have asked to you to design internal controls related to the company's vendor payment system. Remember from your accounting information systems course that there are several important controls in the order-to-pay cycle, including approval of purchases, and approval of vendor invoices. You have implemented the following control:

- The accounting clerk prepares a voucher package containing a disbursement voucher, vendor invoice, purchase order, and receiving document for each payment to be made.
- Invoices where payment exceeds $10,000 have a second level of approval, which is noted in the information system by the treasurer's initials.

You want to test the operational effectiveness of this control by analyzing data captured by the accounting information system during this process. Assume the objective of your analysis is to understand whether the internal control is operating as designed. Design your data analysis strategy to address this objective. Be sure to document the following components:

1. What is the question your data analysis should address?
2. Using the data dictionary, what measured raw data and non-measured raw data fields should you request?
3. What analysis choices will you make to analyze the data? Specifically, what calculated data fields would you create in your analysis?
4. What are the data risks to your data analysis strategy?
5. What are the analysis risks in your strategy?
6. What controls should you include in your data and analysis strategies to minimize the risks you outlined?

PAC 4.2 Auditing: Select Large and Unusual Purchases

Data **Auditing** You are a second-year staff working at a public accounting firm assigned to the ATI engagement, which is a new engagement for the firm. Your audit senior has provided you with a transaction file that contains all the company's payments to vendors for the month of January. You have been asked to determine if the company is accurately recording invoice information by designing a data analysis strategy to select purchase transactions for further testing. You must design a data analysis strategy to select purchase transactions for further testing. Your senior asked you to identify transactions that may be outside the normal purchasing behavior. Therefore, you must perform descriptive analyses. Your senior provided you with this partial data analysis plan.

Complete the chart by identifying the risk and related controls for the data and analysis choices determined in the data analysis plan.

Objective and Questions	Data and Analysis Strategies	Risks	Controls
Objective: Identify purchase transactions for further testing. Questions: Are there purchase transactions that may be considered anomalies?	Data strategy: InvoiceNo, InvoiceDate, QtyInvoice, InvoiceAmt, VendorID, VendorDescription Analysis strategy: Perform descriptive statistics to understand total dollar amount and count of purchases made to each vendor. Perform diagnostic statistics and create a scatterplot whereby the InvoiceDate is on the X-axis and the InvoiceAmount is on the Y-axis to identify any outlier purchases.	**1.** Data: **2.** Analysis:	**3.** Data: **4.** Analysis:

5. Perform descriptive statistical analysis to identify the total dollar amount and count of purchases made to each vendor.
6. Use the provided data set to perform the diagnostic statistical analysis to create a scatterplot where the InvoiceDate is on the X-axis and the InvoiceAmt is on the Y-axis to identify outlier purchases.

PAC 4.3 Financial Accounting: Plan to Identify High Activity Vendors

Financial Accounting The owners of the company want to know which vendors the purchasing group are using and how much they are paying each to negotiate better contract terms with regular vendors. You are performing data analysis to identify the most used vendors. Answer each question with respect to your data analysis.

1. What is the question your data analysis should address?
2. Using the data dictionary, what measured raw data and non-measured raw data fields should you request?
3. What analysis choices will you make to analyze the data? Be specific, what calculated data fields would you create in your analysis?
4. What are the data risks to your data analysis strategy?
5. What are the analysis risks in your strategy?
6. What controls should you include in your analysis to minimize the risks you outlined?

PAC 4.4 Managerial Accounting: Rank Vendor Quality

Data **Managerial Accounting** Because the company is growing so fast, the owners want to be sure that they are partnering with the right vendors for raw materials. They have asked the receiving team to give each receipt a rating to document the quality of the packaging, the materials in appearance, and overall delivery. For each receipt the receiving group documents a quality rating in the accounting information system. The owners have asked you to determine how the quality rating by vendor has changed since the beginning of the year. Your manager has designed the following partial data analysis strategy. Complete the chart and document the risks and related controls to your analyses.

Objective and Questions	Data and Analysis Strategies	Risks	Controls
Objective: Understand the quality ratings of purchases.	Data strategy: VendorID, VendorName, ReceivingNo, ReceivingDate, QualityRate	**1.** Data:	**3.** Data:
Questions: What is the average quality rating by vendor?	Analysis strategy: Calculate the average QualityRating by VendorName.	**2.** Analysis:	**4.** Analysis:
What is the highest and lowest quality rating by vendor?	Calculate the minimum and maximum QualityRating by VendorName.		
How has the quality rating by vendor changed over time?	Visualize how QualityRating by VendorName has changed over time.		

5. Perform the analyses suggested in the data analyses project plan.

PAC 4.5 Tax Accounting: Plan International Duty Cost Analysis

Tax Accounting The owners want to understand the total dollar amount of customs duty they will owe the U.S. government for the year. They have asked you to perform descriptive analytics to understand the purchases from vendors located in each country. Your company makes purchases from vendors located in several countries, such as China, Mexico, the United States, Japan, South Africa, Israel, Greece, and Egypt.

You and your team have designed a data analysis strategy. You have been asked to consider the risks and related controls in the development of that strategy. Complete the following chart.

Objective and Questions	Data and Analysis Strategies	Risks	Controls
Objective: Understand the purchases made from each country.	Data strategy: ShipLocation, InvoiceAmt, FlatDuty, TariffAmt, PONumber	**1.** Data:	**3.** Data:
Questions: What dollar amount of purchases are made from each country?	Analysis strategy: Use a pivot table to drop the ShipLocation into the rows and the sum of InvoiceAmt into values.	**2.** Analysis:	**4.** Analysis:
How many purchases were made from vendors in each country?	Use a pivot table to drop the ShipLocation into the rows and count of PONumber into values.		

Le Grind Continuing Case: Design a Data Analysis Plan for Describing and Diagnosing Gross Margin Changes

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